

DESIGNING NON-MONETARY INCENTIVES FOR EFFICIENT SELFISH ROUTING THROUGH STRATEGIC INTERSECTION CONTROL

A THESIS SUBMITTED TO
THE GRADUATE SCHOOL OF ENGINEERING AND SCIENCE
OF BILKENT UNIVERSITY
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE DEGREE OF
MASTER OF SCIENCE
IN
ELECTRICAL AND ELECTRONICS ENGINEERING

By
Yusuf Saltan
December 2025

Designing Non-Monetary Incentives for Efficient Selfish Routing
through Strategic Intersection Control

By Yusuf Saltan

December 2025

We certify that we have read this thesis and that in our opinion it is fully adequate,
in scope and in quality, as a thesis for the degree of Master of Science.

Muhammed Ömer Sayın(Advisor)

Arif Bülent Özgüler

Saeed Ahmed

Approved for the Graduate School of Engineering and Science:

Orhan Arıkan
Director of the Graduate School

ABSTRACT

DESIGNING NON-MONETARY INCENTIVES FOR EFFICIENT SELFISH ROUTING THROUGH STRATEGIC INTERSECTION CONTROL

Yusuf Saltan

M.S. in Electrical and Electronics Engineering

Advisor: Muhammed Ömer Sayın

December 2025

Urban transportation networks routinely suffer from inefficiencies arising from selfish routing, where individual drivers choose routes that minimize their personal travel time rather than the total system delay. This decentralized behavior leads to suboptimal user equilibria that may substantially deviate from the system-optimal flow. While monetary tolls can theoretically eliminate this inefficiency, their practical, political, and equity limitations motivate the search for alternative, non-monetary mechanisms. This thesis develops and analyzes two complementary intersection-based incentive mechanisms that built on modern Autonomous Intersection Management (AIM) to control delay, influence path costs, and steer selfish routing toward socially optimal patterns without relying on monetary transfers. The first mechanism introduces Strategic Priority-Based Scheduling (SPBS) at intersections, wherein small, route-dependent priority adjustments impose structured waiting-time differences. Analytical examples, including Pigou’s example, demonstrate that even minimal priority asymmetries substantially reduce inefficiency. Microscopic high-fidelity simulations confirm the mechanism’s feasibility under realistic driving and queueing dynamics. The second mechanism generalizes this idea into an analytical framework based on timestamp offsets. Intersections apply small additive adjustments to vehicles’ effective arrival times, inducing path-dependent node delays while preserving monotonicity and separability of cost functions. This structure enables a bilevel optimization formulation in which a system planner designs offsets anticipating user-equilibrium responses. Calibration with simulation-generated intersection delay data for the Sioux Falls network yields realistic quartic node cost models, and large-scale experiments show that timestamp incentives close up to 68 percent of the user-equilibrium inefficiency, even under tight operational constraints.

Together, these mechanisms demonstrate that intersections, traditionally viewed as bottlenecks, can be transformed into powerful, non-monetary control mechanisms. By leveraging modern AIM capabilities, they offer practical, scalable, and analytically grounded tools for improving network-wide efficiency without monetary tolls or major infrastructure changes.

Keywords: Selfish routing, intersection control, non-monetary incentives, price of anarchy, congestion games, mechanism design, autonomous intersection management, bilevel optimization.

ÖZET

STRATEJİK KAVŞAK KONTROLÜ ARACILIĞIYLA VERİMLİ BENCİL YÖNLENDİRME İÇİN PARASAL OLMAYAN TEŞVİK TASARIMI

Yusuf Saltan

Elektrik ve Elektronik Mühendisliği, Yüksek Lisans

Tez Danışmanı: Muhammed Ömer Sayın

Aralık 2025

Kentsel ulaşım ağları, bireysel sürücülerin toplam sistem gecikmesi yerine kendi kişisel seyahat sürelerini en aza indiren rotaları seçtiği bencil yönlendirmeden kaynaklanan verimsizliklerden sıklıkla muzdariptir. Bu merkezi olmayan davranış, sistem-optimal akıştan önemli ölçüde sapabilen, optimal olmayan kullanıcı dengelerine yol açmaktadır. Parasal geçiş ücretleri teorik olarak bu verimsizliği ortadan kaldırılabile de; pratik, politik ve hakkaniyetle ilgili kısıtlamalar, parasal olmayan alternatif mekanizma arayışlarını teşvik etmektedir. Bu tez, parasal transferlere dayanmadan gecikmeyi kontrol etmek, yol maliyetlerini etkilemek ve bencil yönlendirmeyi sosyal açıdan optimal modellere sevk etmek amacıyla modern Otonom Kavşak Yönetimi (AIM) üzerine inşa edilen, birbirini tamamlayıcı iki kavşak tabanlı teşvik mekanizması geliştirmekte ve analiz etmektedir.

İlk mekanizma, küçük ve rotaya bağlı öncelik ayarlamalarının yapılandırılmış bekleme süresi farkları dayattığı, kavşaklarda Stratejik Öncelik Tabanlı Çizelgeleme (SPBS) yöntemini sunmaktadır. Pigou örneğini de içeren analitik örnekler, minimal öncelik asimetrisinin bile verimsizliği önemli ölçüde azalttığını göstermektedir. Mikroskobik yüksek sadakatli simülasyonlar, gerçekçi sürüş ve kuyruk dinamikleri altında mekanizmanın uygulanabilirliğini doğrulamaktadır. İkinci mekanizma ise bu fıkri zaman damgası ofsetlerine dayalı analitik bir çerçeveye genelleştirmektedir. Kavşaklar, araçların efektif varış zamanlarına küçük toplamsal ayarlamalar uygulayarak, maliyet fonksiyonlarının monotonluğunu ve ayrılabilirliğini korurken yola bağlı düğüm gecikmeleri oluşturur. Bu yapı, bir sistem planlayıcısının kullanıcı dengesi tepkilerini öngörerek ofsetleri tasarladığı iki seviyeli bir optimizasyon formülasyonuna olanak tanır. Sioux Falls ağı için simülasyonla üretilen kavşak gecikme verileriyle yapılan kalibrasyon,

gerçekçi dördüncü dereceden düğüm maliyet modelleri sunmakta ve büyük ölçekli deneyler, zaman damgası teşviklerinin sıkı operasyonel kısıtlar altında bile kullanıcı dengesi verimsizliğinin yüzde 68'ine kadarını giderdiğini göstermektedir.

Birlikte ele alındığında bu mekanizmalar; geleneksel olarak darboğaz şeklinde görülen kavşakların güçlü, parasal olmayan kontrol mekanizmalarına dönüştürülebileceğini göstermektedir. Bu yöntemler, modern AIM yeteneklerinden yararlanarak, parasal geçiş ücretleri veya büyük altyapı değişiklikleri olmaksızın ağ genelinde verimliliği artırmak için pratik, ölçeklenebilir ve analitik temelli araçlar sunmaktadır.

Anahtar sözcükler: Bencil yönlendirme, kavşak kontrolü, parasal olmayan teşvikler, anarşinin bedeli, tıkanıklık oyunları, mekanizma tasarımı, otonom kavşak yönetimi, iki seviyeli optimizasyon.

Acknowledgement

I would like to express my sincere gratitude to my advisor, Asst. Prof. Muhammed Ömer Sayın, for his guidance and encouragement, throughout this research. His insights and perspective were invaluable in shaping this thesis.

I would also like to thank my lab colleagues and friends, Yüksel, Arda, Taha, and Ahmed Said, for the stimulating discussions, collaborative environment, and friendship that made this work both productive and enjoyable.

This thesis is supported by the Scientific and Technological Research Council of Türkiye (TUBITAK) BİDEB 2232-B International Fellowship for Early Stage Researchers under Grant Number 121C124.

Contents

1	Introduction	1
2	Literature Review	7
2.1	Traffic Congestion and Selfish Routing	7
2.2	Nonatomic Congestion Games and Wardrop Equilibrium	9
2.3	Price of Anarchy and Inefficiency in Selfish Routing	10
2.4	Incentive Mechanisms for Efficient Routing	11
2.5	Intersection Control in Traffic Networks	15
2.6	Autonomous Intersection Management (AIM)	19
2.7	Game-Theoretic Models of Intersection Control and Priority Systems	22
2.8	Mechanism Design Perspective and the Bridge to Timestamp- Based Intersection Incentives	26
3	Strategic Control of Intersections for Efficient Traffic Routing without Tolls	30
3.1	Preliminaries: Routing on Pigou’s Network	31

3.2	Autonomous Intersection Management and FCFS Baseline	33
3.3	Strategic Priority-Based Scheduling (SPBS)	34
3.4	Qualitative Cost Comparison: FCFS vs. SPBS	36
3.5	Analytical Characterization of SPBS	37
3.6	Microscopic Simulation Study in SUMO	39
3.7	Discussion and Takeaways	42
4	Designing Non-monetary Intersection Control Mechanisms for Efficient Selfish Routing	43
4.1	Overview	43
4.2	Proposed Two-Layer Intersection Control Mechanism	46
4.3	Problem Formulation	49
4.3.1	Analytical Framework	50
4.4	Equilibrium Characterization	54
4.5	Bilevel Optimization Formulation	56
4.6	Illustrative Examples	58
4.6.1	Braess Network	58
4.6.2	Experimental Analyses on the Sioux Falls Network	61
5	Conclusion	66
5.1	Main Findings and Contributions	67

5.2	Limitations	68
5.3	Future Research Directions	69
5.4	Closing Remarks	69
A	Proof of Proposition 4.9	70
B	Proof of Theorem 4.10	72
C	Experimental Setup and Reproducibility	74

List of Figures

1.1	Before (a) and after (b) adding the zero-cost edge.	2
1.2	Autonomous intersection management enabled by vehicle-to-infrastructure communication. Vehicles transmit identification and request information to RSUs, which schedule crossing times while respecting safety constraints.	3
3.1	Pigou’s example.	31
3.2	Illustrative autonomous intersection management architecture. A roadside unit receives requests from approaching vehicles and schedules their passage through the intersection.	33
3.3	Qualitative comparison of FCFS and SPBS on Pigou’s network. The top row shows how FCFS produces symmetric node delays and a UE fully on the lower route. The bottom row illustrates how SPBS introduces route-dependent node delays that shift the equilibrium toward the upper route and reduce total cost. Shaded regions represent total travel cost at equilibrium for each route.	36
3.4	Average travel cost (edge + node) for upper and lower routes under FCFS and SPBS from SUMO simulations. The qualitative behavior is consistent with the analytical node-cost model used for SPBS.	41

4.1	Two-layer mechanism: vehicles submit requests with timestamps, the system planner applies timestamp adjustments to generate non-monetary incentives, and RSUs schedule intersection usage based on the modified timestamps.	48
4.2	Illustration of path costs under the optimal flow f^* in the absence of incentives. Timestamp-based adjustments can raise or lower path costs to ensure that f^* satisfies the Wardrop equilibrium inequalities.	52
4.3	Comparison of the original (a) three-node/four-edge network and (b) the transformed version with node splitting. The path $p_1 = se_1ve_3t$ and its counterpart in the transformed network are highlighted in red.	55
4.4	A Braess network.	59
4.5	Microscopic–macroscopic linkage: a representative intersection simulated in SUMO (left) and the full Sioux Falls network (right).	61
4.6	A plot of average travel time (second) versus flow rate (vehicles per second) within the control zone and the flow rate for the simulated Node 10 using the FCFS protocol under no, positive and negative timestamp modifications for one of the paths.	63
4.7	Convergence of social cost under SPSA optimization for different incentive bounds.	64
4.8	Comparison of network flow distributions in the Sioux Falls benchmark. (a) Baseline user–equilibrium (UE) showing congestion concentration on central corridors. (b) Flow redistribution after timestamp-based incentives, where red (and blue) edges depict increased (and decreased) flows.	65

List of Tables

4.1	Equilibrium flows and social costs in the Braess network under two cost models.	59
4.2	Equilibria and social costs in the Braess network under two cost models.	60
C.1	Fitted quartic node cost coefficients from SUMO simulations for the 24 Sioux Falls intersections. Quartic models follow $c_v(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$. Columns list coefficients (a_0 – a_4) and fitting error (MAPE, %). Rows correspond to the 24 nodes. . . .	76

Chapter 1

Introduction

Urban transportation networks face growing pressure from increasing travel demand, limited infrastructure capacity, and the decentralized nature of routing choices. As cities expand and vehicle usage intensifies, congestion has become an unavoidable feature of daily mobility. The resulting delays impose substantial social costs in the form of lost productivity, fuel waste, greenhouse-gas emissions, and increased accident risk [1, 2]. While traditional congestion-management strategies focus on road expansion, signal design, or tolling, the fundamental challenge remains: drivers act selfishly, selecting the routes that minimize their own travel time, often at the expense of system-wide efficiency.

This misalignment between individual and collective objectives is central to the theory of selfish routing. When each driver chooses the perceived fastest route, traffic flows converge to a *Wardrop equilibrium*, which need not coincide with the *system-optimal* flow that minimizes total travel time [3, 4]. The inefficiency induced by this decentralized behavior is captured by the *Price of Anarchy* (PoA), which quantifies how much worse equilibrium performance can be compared to optimal routing [5]. Empirical and theoretical results show that PoA values can be significant, particularly in congested urban networks where drivers repeatedly overload a small number of apparently attractive paths.

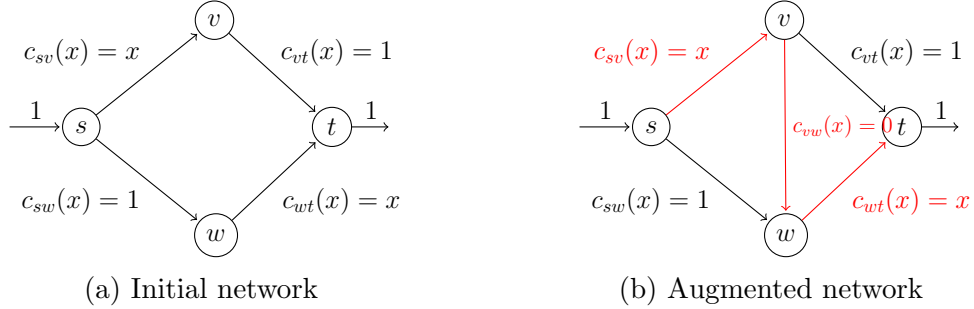


Figure 1.1: Before (a) and after (b) adding the zero-cost edge.

To illustrate how selfish routing can generate counterintuitive and inefficient outcomes, consider the classical *Braess' paradox*. Suppose a unit mass of drivers must travel from s to t using one of two symmetric routes, as shown in Fig. 1.1a. Each edge (a, b) has an associated latency $c_{ab}(x)$ that depends on the fraction x of traffic using that edge. In the initial network, the unique *user equilibrium*, a state where no driver can reduce their travel time by unilaterally changing routes, occurs when the flow splits evenly across the two paths $s \rightarrow v \rightarrow t$ and $s \rightarrow w \rightarrow t$. Under this equilibrium, they have the travel cost of 1.5. Now suppose a new, seemingly beneficial zero-cost link between v and w is added, creating the augmented network in Fig. 1.1b. At equilibrium in this augmented network, *all* flow shifts to the path $s \rightarrow v \rightarrow w \rightarrow t$, as each driver attempts to exploit the shortcut. However, the resulting congestion on the flow-dependent edges increases the travel cost to 2, which is strictly worse than in the original network.

This example highlights a fundamental challenge: inefficiency in routing is not solely the consequence of insufficient infrastructure, but rather the result of strategic behavior under decentralized decision-making. It motivates the need for mechanisms capable of guiding drivers toward more efficient flows without relying exclusively on capacity expansion or monetary tolls.

Another key observation is that *intersections*, rather than road segments, are often the primary contributors to urban congestion. Intersection delays arise from conflict resolution between crossing movements, signal phasing, and queueing dynamics. These node-level delays frequently dominate overall travel time

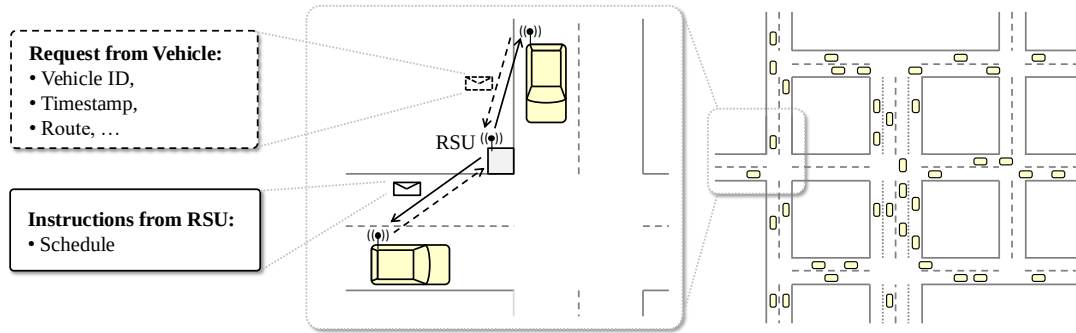


Figure 1.2: Autonomous intersection management enabled by vehicle-to-infrastructure communication. Vehicles transmit identification and request information to RSUs, which schedule crossing times while respecting safety constraints.

in dense networks, yet most congestion-game analyses treat intersections implicitly or aggregate their effects into edge costs. In reality, intersections provide a natural and powerful means of influencing driver behavior: delays at junctions are unavoidable, non-monetary, and inherently linked to operational rules such as priorities, reservations, and scheduling policies.

Recent advances in connected and autonomous vehicle (CAV) technologies, vehicle-to-infrastructure (V2I) communication, and Autonomous Intersection Management (AIM) create new opportunities for fine-grained intersection control. Modern AIM systems can regulate the order, timing, and priority of vehicles entering conflict zones, enabling intersection managers to adjust waiting times with high temporal resolution, (see Fig. 1.2). This capability suggests a compelling question:

Can strategic and fine-grained autonomous intersection management reduce routing inefficiencies by steering self-interested drivers toward socially optimal and equitable traffic patterns?

This thesis answers this question affirmatively by developing two complementary classes of intersection-based incentive mechanisms. Both mechanisms exploit

the inherent delay-generating structure of intersections, but they operate at different levels of abstraction and control.

Motivation. Monetary mechanisms such as tolls, congestion pricing, or tradable credits are effective in theory but face substantial obstacles in practice: equity concerns, political rejection, enforcement complexity, and technological requirements. In contrast, *intersection delays are inherently non-monetary* and require no external payment systems. Because intersections already impose waiting time, modifying this delay, even slightly, offers a feasible and socially acceptable means of influencing driver behavior.

Furthermore, AIM systems provide the microscopic control necessary to implement such mechanisms: vehicles can be assigned different priorities, timestamps, or service orders depending on their intended movements. These features enable *path-dependent* delay adjustments that effectively reshape perceived travel costs. When incorporated into a congestion-game framework, these adjustments steer equilibrium flows toward more efficient patterns without monetary transfers.

Two Mechanisms, One Objective. This thesis unifies two research lines, corresponding to the two included papers, that address the same fundamental question through distinct approaches.

Mechanism 1: Strategic Priority-Based Scheduling

The first mechanism investigates how *local priority rules* at intersections can reduce inefficiency caused by selfish routing. Using simple but illustrative networks such as Pigou’s example, the mechanism assigns vehicles to “priority” or “non-priority” classes based on their chosen routes. Priority vehicles receive faster service, and non-priority vehicles experience limited additional delay.

This approach demonstrates that even small, locally administered priority adjustments are sufficient to shift user-equilibrium flows toward significantly more efficient patterns. By influencing route choice through non-monetary priority

modifications, the mechanism reduces overall congestion and narrows the gap between equilibrium and system-optimal performance. Moreover, because these incentives can be implemented directly through existing or emerging intersection-control technologies, the framework provides a practical and deployable pathway for improving network efficiency without relying on monetary tolls or major infrastructural changes.

The analysis combines theoretical results, equilibrium computations, and microscopic SUMO simulations to show how local intersection policies influence global routing outcomes.

Mechanism 2: Timestamp-Based Node Costs and Bilevel Optimization

The second mechanism extends the conceptual framework significantly. Instead of assigning categorical priority classes, intersections apply *timestamp offsets*, namely small, continuous adjustments to vehicle processing time based on their chosen paths. These adjustments translate into path-dependent node costs, enabling fine-grained shaping of perceived travel times.

This mechanism exhibits several important properties. First, it extends delay-based incentives into a mathematically rigorous framework in which node costs become flow-dependent and path-specific. Second, the resulting cost structure preserves convexity and admits a potential formulation, guaranteeing the existence and essential uniqueness of the induced equilibrium. Third, the mechanism naturally yields a bilevel optimization problem in which a system operator selects timestamp offsets while explicitly accounting for the selfish equilibrium response of drivers. Finally, its practical validity is demonstrated through microscopic SUMO experiments and network-level analyses on the Sioux Falls benchmark, confirming that the theoretical model captures realistic intersection behavior and produces meaningful efficiency improvements.

Compared to the first mechanism, the timestamp-based approach provides a

scalable and analytically grounded method for network-wide optimization.

Unified Thesis Contributions. Together, the two mechanisms developed in this thesis position intersections as effective, practical levers for influencing routing behavior in selfish transportation networks. By treating intersection delays as controllable, non-monetary incentives, the thesis establishes a unified framework that links local priority decisions to global traffic efficiency. The first mechanism shows, through theoretical analysis and simulation, that even simple priority-based scheduling rules can meaningfully shift equilibrium flows. The second mechanism generalizes this idea by introducing timestamp-based, path-dependent node costs that integrate seamlessly into a convex potential framework and support a bilevel optimization formulation. Across both approaches, the thesis demonstrates the existence of well-behaved equilibria, quantifies the resulting efficiency gains, and validates the mechanisms on networks ranging from stylized examples to realistic urban topologies. Collectively, these contributions offer a rigorous and practically implementable foundation for non-monetary congestion management using emerging V2I and autonomous intersection technologies.

Thesis Organization

Chapter 2 gives a general literature review for the background. Chapter 3 presents the priority-based scheduling mechanism and its analytical insights. Chapter 4 develops the timestamp-based node-cost mechanism and bilevel optimization formulation, including simulation-based calibration. Chapter 5 concludes with a unified discussion of findings, limitations, and directions for future research.

Together, these chapters demonstrate that intersections, traditionally viewed as an unavoidable bottlenecks, can be transformed into powerful, fine-grained incentive control mechanisms that guide self-interested drivers toward socially efficient routing, without requiring monetary payments or major infrastructure changes.

Chapter 2

Literature Review

2.1 Traffic Congestion and Selfish Routing

Urban transportation networks face persistent congestion arising from growing vehicle demand, limited infrastructure, and the decentralized nature of routing decisions. The resulting delays impose substantial social costs, including fuel consumption, productivity loss, increased accident risk, and elevated greenhouse-gas emissions [1, 2]. As metropolitan regions continue to densify, congestion increasingly becomes a system-wide, multi-period phenomenon rather than an isolated peak-hour issue, making its analysis intrinsically complex.

A fundamental feature of road traffic is that route choice is *self-organized*: drivers select paths that minimize their own travel time based on prevailing conditions, individual perception, or navigation applications. While rational from the perspective of each driver, this behavior can be detrimental at the network level. Selfish-routing models formalize this tension by treating each driver as an individual optimizer embedded within a larger population. When many drivers behave in this way, they may overload “attractive” paths and create congestion externalities that degrade performance for all users [4].

Two classical examples illustrate these inefficiencies. The first is *Pigou’s example* [6], a two-link network in which all drivers gravitate toward the link with flow-dependent delay, producing worse outcomes than if traffic were split optimally. This simple construction reveals that individual incentives need not align with societal interest. A more striking phenomenon, *Braess’s paradox* [7], shows that adding an apparently beneficial road can increase congestion at equilibrium. Conversely, network performance may improve if certain links are removed. These paradoxes underscore that infrastructure enhancements alone cannot ensure efficient operation when drivers act selfishly.

The divergence between equilibrium and socially optimal flows is often substantial in large-scale networks [3, 5]. The *Price of Anarchy* (PoA) is widely used to quantify this gap and is known to be significant in both theoretical and empirical studies. This motivates the development of *mechanism design* approaches that nudge or guide drivers toward more efficient routing patterns. Traditional approaches such as monetary tolling [8, 9] are effective in principle but face political, equity, or practical constraints. As autonomous intersection management becomes more prevalent, non-monetary mechanisms, particularly those implemented at intersections, emerge as viable alternatives.

To analyze selfish routing formally and design mechanisms that mitigate its inefficiency, we adopt the framework of *nonatomic congestion games*. The next section introduces this model and the Wardrop equilibrium concept that describes decentralized route selection.

2.2 Nonatomic Congestion Games and Wardrop Equilibrium

Nonatomic congestion games provide a foundational mathematical framework for modeling selfish routing. In this paradigm, traffic consists of a continuum of infinitesimal agents, each of whom controls a negligible portion of overall demand. While no single driver can meaningfully influence congestion, the collective decisions of all drivers shape network-wide delays.

Consider a directed graph $G = (V, E)$ representing the transportation network. Let $\mathcal{P}$ denote the set of all simple paths between a given origin–destination (OD) pair, and let $f = (f_p)_{p \in \mathcal{P}}$ denote the path-flow vector satisfying

$$\sum_{p \in \mathcal{P}} f_p = 1, \quad f_p \geq 0,$$

where f_p represents the fraction of traffic choosing path p . The induced flow on an edge is

$$f_e = \sum_{p \in \mathcal{P}: e \in p} f_p.$$

Each edge e has a latency function $c_e(f_e)$ that maps flow to average travel time. These functions are typically continuous, non-negative, and non-decreasing, reflecting the intuitive relationship between congestion and delay. Popular choices include affine latency functions and the Bureau of Public Roads (BPR) model [10]. Path costs follow additively:

$$c_p(f) = \sum_{e \in p} c_e(f_e).$$

When drivers independently choose minimum-cost paths, the system reaches a *Wardrop equilibrium* [3].

Definition 2.1 (Wardrop Equilibrium). A feasible flow f^* is a Wardrop equilibrium if every used path p satisfies

$$c_p(f^*) \leq c_{p'}(f^*) \quad \forall p' \in \mathcal{P}.$$

A key result due to Beckmann *et al.* [11] is that if latency functions are non-decreasing, Wardrop equilibria coincide with the minimizers of the convex potential function

$$\Phi(f) = \sum_{e \in E} \int_0^{f_e} c_e(z) dz.$$

Thus equilibrium computation becomes a convex optimization problem, enabling tractable numerical methods such as the Frank–Wolfe algorithm.

Despite its stability, Wardrop equilibrium seldom matches the system-optimal flow, which minimizes the total latency

$$C(f) = \sum_{p \in \mathcal{P}} f_p c_p(f) = \sum_{e \in E} f_e c_e(f_e).$$

This disparity reflects uninternalized congestion externalities: each driver accounts only for their own travel time, not the delays imposed on others. The next section examines this inefficiency through the concept of the Price of Anarchy.

2.3 Price of Anarchy and Inefficiency in Selfish Routing

The *Price of Anarchy* (PoA) quantifies the degradation in network performance caused by selfish routing. Defined as the ratio

$$\text{PoA} = \frac{C(f^{\text{WE}})}{C(f^{\text{SO}})},$$

it measures how much worse equilibrium is compared to the best possible (system-optimal) traffic allocation [12, 5]. A PoA greater than one indicates inefficiency, and; while theoretical bounds can reach 2.15 for standard delay functions, values between 1.1 and 1.3 are more commonly observed in empirical studies of major cities [13, 14].

Theoretical results show that PoA depends on latency function structure. For affine latencies, Roughgarden and Tardos [5] proved the tight bound $\text{PoA} = 4/3$.

For polynomial latencies of degree n , the worst-case PoA grows as

$$\Theta\left(\frac{(n+1)^{n+1}}{n^n}\right),$$

which for the BPR model ($n = 4$) yields a PoA exceeding 2. These results reveal that strategic behavior can have significant adverse effects, particularly in congested settings.

Paradoxical phenomena further highlight structural sources of inefficiency. In Pigou-like networks, selfish routing leads to overuse of congestible links [6]. In Braess networks, adding a link worsens equilibrium travel times [7, 15]. These examples illustrate how network topology and latency interactions influence equilibrium outcomes.

Empirical studies and large-scale simulators such as SUMO, MATSim, and Aimsun corroborate these theoretical findings: equilibrium flows often overload central arterials while peripheral routes remain underutilized [16, 17]. Since monetary tolls face political and practical barriers, attention has shifted toward non-monetary mechanisms to counteract equilibrium inefficiencies and guide traffic toward more efficient flow patterns.

2.4 Incentive Mechanisms for Efficient Routing

Persistent inefficiencies of Wardrop equilibria have motivated a wide range of approaches designed to steer traffic toward socially desirable operating points. These mechanisms vary widely in their assumptions, infrastructure requirements, and means of influencing driver behavior. This section surveys the principal categories of incentive mechanisms proposed in the literature, including monetary tools, tradable and non-monetary systems, information-based strategies, infrastructure modification, fairness-oriented designs, and intersection-level control to provide context for the timestamp-based mechanism developed later in this thesis.

Monetary Incentives: Tolls and Marginal-Cost Pricing

A classical approach to reducing congestion is the use of *Pigouvian tolls*, which charge drivers the marginal external cost they impose on others [6, 18]. If implemented exactly, such tolls align user equilibrium with the system-optimal flow [11]. Extensive research examines marginal-cost tolls, dynamic tolling, and pricing schemes under various network and behavioral models [9, 19, 20]. However, practical deployment faces significant challenges. First, there are substantial **equity and political concerns**: low-income drivers may be disproportionately burdened [21, 22], leading to public resistance. Second, tolling requires detailed and accurate **information on marginal delay**, which depends on sophisticated traffic modeling capable of capturing real-time conditions. Third, implementation demands considerable **technological infrastructure**, including sensing, billing, and enforcement systems capable of supporting real-time pricing.

While deployments in cities such as London, Stockholm, and Singapore [23, 24] demonstrate that tolling can reduce congestion, these political, informational, and technological barriers limit its broader adoption. Consequently, these limitations motivate the exploration of non-monetary alternatives.

Tradable Credits and Artificial Tokens

To mitigate distributional concerns, *tradable credit schemes* propose allocating each driver a budget of congestion credits, which may be spent or exchanged rather than paid monetarily [25, 26]. These schemes create market-based incentives similar to tolls but avoid direct charges, and related ideas, such as artificial currencies or token-based coordination mechanisms [27], extend naturally to multi-agent systems, including transportation.

Despite their conceptual appeal, tradable credits face several adoption barriers. They require dedicated infrastructure for distributing and trading credits, and the tokens themselves may lack value outside the transportation system [28], making

user acceptance difficult. In addition, the need for operational oversight increases administrative complexity. Together, these factors limit the practicality of large-scale deployment.

Information-Based Mechanisms

Information design influences route choice by shaping drivers’ beliefs rather than altering travel-time functions. Using ideas from Bayesian persuasion [29], central planners can partially or strategically reveal traffic information to distribute demand across routes [30, 31]. This approach, however, faces several limitations. Competing navigation services often disclose real-time traffic conditions [32], reducing the planner’s ability to influence beliefs. Drivers may also distrust recommendations when they contradict personal experience, and personalized routing advice introduces privacy concerns. Despite these challenges, information-based routing remains an active area of research, particularly as predictive models continue to improve [33] and move beyond point estimates toward learning full travel-time distributions [34].

Infrastructure and Network Modifications

Physical modifications such as adding or removing links, adjusting capacities, or optimizing signal timing can significantly affect equilibrium routing [35, 36, 37]. However, structural paradoxes (e.g., Braess’s paradox [7]) highlight that additional capacity may worsen performance. Also network design is typically NP-hard [38, 39], requiring anticipatory modeling of equilibria.

Real-world interventions such as road diets or exclusive bus lanes demonstrate that operational or geometric adjustments can meaningfully reshape flows [40, 41].

Fairness-Constrained Routing Mechanisms

Routing interventions may raise distributional concerns, motivating fairness-aware formulations. Common approaches include bounding increases in individual travel times [42], equalizing travel-time ratios across groups [43], and ensuring Pareto or near-Pareto improvements where no user is worse off [44]. Such constraints are essential for policy acceptance but introduce additional complexity and can limit achievable efficiency improvements.

Intersection-Level Control and Priority Mechanisms

While many methods target route-level incentives, intersections are often the dominant sources of delay [45]. Advances in autonomous intersection management (AIM) enable precise, reservation-based scheduling [46], conflict-graph-based control [47], and dynamic priority rules for transit or emergency vehicles [48, 49]. These systems primarily optimize local intersection performance but increasingly reveal the potential for network-wide influence.

Recent studies show that deliberately modifying intersection waiting times via priority scheduling, timestamp adjustments, or virtual queues can introduce path-dependent incentives without monetary payments [50]. Such *non-monetary* mechanisms leverage the natural delays at intersections to influence route choice in a controlled and operationally feasible manner.

Summary

Existing incentive mechanisms span monetary, market-based, informational, and infrastructural approaches. Monetary tolls provide strong theoretical guarantees but remain politically difficult. Tradable credits reduce equity concerns but require administrative overhead. Information design can influence behavior but competes with real-time navigation services. Infrastructure modifications offer

structural improvements but involve high cost and uncertainty.

Intersection-level control bridges these gaps by offering a non-monetary, operationally grounded means of adjusting perceived path costs. Priority manipulation and timestamp-based scheduling form a promising class of mechanisms capable of influencing equilibrium behavior without additional monetary systems or large-scale construction. These ideas motivate the timestamp-based incentive framework developed in the following chapters.

2.5 Intersection Control in Traffic Networks

While much of the congestion-game literature focuses on route choice and link flows, the performance of urban transportation systems is often dominated by the behavior of intersections. At signalized junctions, competing movements must be coordinated through phasing, priority rules, and physical constraints. Poorly timed or overloaded intersections can degrade the performance of otherwise well-designed networks, whereas effective control strategies can significantly reduce delays even under heavy demand [45, 51]. This section reviews classical and modern approaches to intersection control, emphasizing how node-level delays interact with network-wide routing.

Intersections as Bottlenecks and the Role of Node Delays

Intersections serve as natural bottlenecks because conflicting traffic streams must be separated in time. Even if upstream links have sufficient capacity, queues can form due to signal phasing, pedestrian crossings, turning movements, and limited discharge rates [52]. As a result, intersection delay often constitutes a major portion of total travel time in dense urban environments.

Classical macroscopic models treat these delays as *node costs*, typically approximated through analytical formulas derived from queueing theory. Webster’s

delay formula [53] and its extensions estimate average delay as a function of cycle length, green time, and degree of saturation. The Highway Capacity Manual (HCM) [52] provides more detailed formulations that incorporate stochastic arrivals and multi-phase control. Although these models are widely used for planning, early routing analyses rarely integrated intersection delays explicitly, often treating them implicitly or as aggregated constants. With the rise of advanced signal control methods, the interaction between node-level delays and path choice has become increasingly important.

Fixed-Time Signal Control

The most traditional form of intersection control is *fixed-time* operation, in which signal timings are determined offline and repeat cyclically. Optimization frameworks such as those of Allsop [54] or arterial “green wave” coordination strategies [55] aim to minimize expected delay under historical traffic conditions.

Fixed-time control is simple and robust under moderate variability but cannot adapt to short-term or unexpected fluctuations in demand. Consequently, performance degrades when actual flows deviate from the conditions underlying the design of timing plans.

Actuated and Semi-Actuated Control

To enhance responsiveness, *actuated* and *semi-actuated* controllers use real-time detector inputs (loops, radar, video) to adjust phase timings dynamically [51]. Fully actuated control initiates and terminates phases based on demand on all approaches, whereas semi-actuated control actuates only minor streets while protecting fixed timing for major flows.

These systems reduce delay at isolated intersections with variable demand but are less effective when many intersections interact closely. Purely local actuation may produce miscoordination or oscillations in dense networks, motivating the

need for coordinated or adaptive approaches.

Adaptive and Coordinated Signal Control

Adaptive signal control extends actuation by continuously updating timing plans based on network-level observations. Representative systems include SCOOT [56], which adjusts splits, cycle lengths, and offsets using detector data in real time, and SCATS [57], which selects timing plans from a library based on measured volume and occupancy. Furthermore, systems like RHODES [58] employ hierarchical rolling-horizon optimization to determine phase sequences. Field deployments show that adaptive systems reduce delay on arterial corridors and grid networks, particularly under time-varying demand [59]. In modeling terms, adaptive control implies that node delay functions depend dynamically on measured flows, but most planning tools still treat routing and control as separate modules, complicating the analysis of how signal policies shape selfish routing behavior.

Max-Pressure and Model Predictive Control

Recent developments in control theory have introduced *max-pressure* policies and model predictive control (MPC) schemes that explicitly optimize intersection performance. Max-pressure control selects the phase maximizing a pressure metric based on queue lengths [60, 61], achieving provably stable and throughput-optimal performance under certain assumptions. MPC formulations specify an optimization problem over a prediction horizon that enforces capacity, phase-transition, and queue constraints [62, 63].

These advanced controllers treat intersections as dynamic servers with time-varying service rates. However, integrating them with real-time routing decisions remains challenging, especially when drivers adapt strategically or when congestion spreads across multiple interacting intersections.

Queueing-Theoretic and Analytical Delay Models

Analytical delay models are essential for transportation planning and for linking intersection control to network performance. Queueing models (M/M/1, M/D/1) and gap-acceptance theory form the basis of many delay formulas used in macroscopic simulators and in the HCM [64, 65]. These models provide computationally light approximations but may overlook stochastic spillbacks or interactions across intersections.

Microscopic and agent-based simulation tools such as SUMO, VISSIM, and MATSim provide high-fidelity representations of vehicle trajectories, queue dynamics, and signal operations [66, 16, 17]. They allow realistic calibration of node costs and are therefore crucial for analyses that seek to link intersection-level control to network-wide routing behavior, as undertaken in this thesis.

Discussion and Link to Incentive Design

Intersection control has evolved from simple fixed-time plans to adaptive, coordinated, and optimization-based systems. Yet most approaches focus on local metrics such as delay reduction, queue stability, or throughput without explicitly accounting for strategic routing responses at the network level.

As V2I communication and semi-autonomous driving technologies mature, intersections gain the ability to influence individual vehicle delays in fine-grained, path-dependent ways. This creates an opportunity to integrate intersection control with mechanism design: by deliberately modulating node-level delays, system operators can steer route choices and mitigate the inefficiencies of selfish routing. The timestamp-based incentive framework developed later in this thesis builds upon this perspective, using intersection delays as controlled, non-monetary signals that shape equilibrium flows across the entire network.

2.6 Autonomous Intersection Management (AIM)

The emergence of connected and autonomous vehicles has created opportunities to manage intersections with far greater temporal and spatial precision than is possible with traditional traffic signals. Autonomous Intersection Management (AIM) systems leverage vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication to coordinate vehicle movements at a microscopic level, treating intersections as shared resources whose access must be scheduled safely and efficiently [46, 50]. This section reviews the foundational concepts, scheduling algorithms, safety guarantees, communication requirements, and deployment challenges associated with AIM, with a focus on the aspects that motivate intersection-level incentive design.

Motivation and Foundations

Conventional signal control partitions time into phases that separate conflicting movements, often introducing idle periods and inefficient utilization during low or imbalanced demand. AIM addresses these limitations by dynamically coordinating individual vehicles or platoons based on their positions, velocities, and desired trajectories [67, 68]. Potential benefits include reduced delays, increased throughput, smoother trajectories, and improved safety through automated conflict resolution [69]. Achieving these gains relies on reliable communication (DSRC, C-V2X, or 5G), accurate localization, and responsive control architectures [70, 71].

Reservation-Based Approaches

The seminal reservation-based AIM framework of Dresner and Stone, [46], treats the intersection as a resource that vehicles access by submitting space-time reservation requests. The intersection manager evaluates conflicts with already granted reservations, originally using a grid of reservation tiles, and accepts or rejects requests accordingly.

Extensions to this framework include enhanced performance analyses of tile-based systems [72], conflict-point representations that enable rigorous mixed-integer programming formulations [73], and specialized reservation mechanisms for vehicle platoons [74]. While conceptually simple, reservation-based systems must contend with computational challenges under heavy traffic, motivating the ongoing development of efficient conflict-detection heuristics.

Scheduling Models and Conflict Graphs

A large class of AIM methods recasts intersection management as a scheduling problem over a conflict graph. Each vehicle is represented as a task requiring exclusive use of particular conflict zones or resources. This abstraction enables classical scheduling techniques, including First-Come-First-Served (FCFS) for fairness, [46], and priority-based scheduling for transit or emergency vehicles [41, 49]. Furthermore, approaches such as backpressure or max-weight scheduling, [60], draw inspiration from queueing networks, while graph-coloring or ordering algorithms facilitate conflict-free sequence generation [75, 73]. These methods offer interpretability, flexibility, and compatibility with priority structures which are key properties for designing path-dependent incentives.

Communication and Reliability Requirements

AIM performance depends critically on communication reliability, with technologies varying in latency, bandwidth, and range. DSRC systems rely on IEEE 802.11p with moderate latency [70], whereas C-V2X and 5G provide enhanced capacity for dense traffic environments [71]. Communication failures, packet loss, or synchronization errors may compromise scheduling accuracy or safety, necessitating fallback strategies such as reverting to signal control or locally safe default behaviors [76].

Safety Guarantees

Safety remains the central requirement of AIM systems, where formal verification tools and control-theoretic methods provide provable guarantees. For instance, control barrier functions define invariant sets to prevent collisions [77], while reachability analysis generates safe forward-in-time sets even under disturbances [78]. Additionally, hybrid system models effectively capture discrete transitions, such as entering or exiting conflict zones, to design supervisory controllers [79]. These methods typically constrain or validate the scheduling layer rather than replace it entirely, ensuring AIM operates safely even under uncertainty.

Learning-Based Approaches

Reinforcement learning (RL) and multi-agent learning have recently been applied to AIM, enabling data-driven policies for scheduling or trajectory planning [80, 81]. Centralized approaches optimize intersection-wide policies, while decentralized schemes assign individual controllers to vehicles or subregions [82]. Hierarchical RL blends high-level scheduling with low-level safety-constrained controllers. These methods show promise but face challenges related to safety certification, generalization, and interpretability; hybrid architectures combining RL with rule-based constraints are increasingly favored [83].

Deployment Challenges

Despite substantial progress, widespread AIM deployment faces practical constraints: mixed autonomy with human-driven vehicles [84], the need for fail-safe operation under communication loss, regulatory and liability issues, and computational scalability in dense traffic. Early adoption is therefore expected in hybrid or partial-AIM systems, such as vehicle-platooning support or priority-based reservation for transit and emergency vehicles.

Intersection Control as a Basis for Incentives

A critical feature of AIM is its ability to manipulate individual vehicle delays directly through reservation ordering, priority rules, or timestamp adjustments. Unlike monetary tolls, such delay-based mechanisms modify perceived travel times without financial transactions. This makes AIM an ideal foundation for designing non-monetary, path-dependent incentives. The timestamp-based mechanism developed in this thesis leverages precisely this capability: by adjusting vehicle service times at intersections in a controlled and path-specific manner, the system influences route choice while preserving safety and operational feasibility.

2.7 Game-Theoretic Models of Intersection Control and Priority Systems

As intersections increasingly incorporate communication technologies and dynamic control systems, driver interactions at these nodes become inherently strategic. Rather than passively obeying fixed signal rules, modern vehicles—especially connected or autonomous ones—may anticipate or influence queue positions, scheduling decisions, or priority allocations. Game-theoretic models provide the tools to analyze such behavior. This section reviews strategic interactions at intersections, focusing on priority competition, queueing games, scheduling games, auction-based schemes, coalition formation, information asymmetry, and mechanism design approaches that explicitly account for strategic behavior.

Priority Rules and Strategic Queueing

Intersections can be modeled as queueing systems where drivers compete for service under rules such as first-come-first-served (FCFS), yield protocols, or stop-sign priority. Queueing-game analyses show that drivers may strategically adjust

arrival times or speeds to secure favorable queue positions [85, 86]. Classic results highlight that under FCFS, strategic “arrival rushing” can create excessive peak load [87]. Furthermore, priority rules can generate externalities, as speed changes by one group alter delays for others. In mixed-autonomy settings, autonomous vehicles may exploit superior information or predictability to gain priority over human drivers [88]. These models illustrate the incentives users confront even under deterministic service rules.

Intersection Scheduling as a Strategic Game

More detailed AIM systems treat intersections as scheduling platforms where vehicles request trajectories, arrival times, or space–time reservations, and conflicts must be resolved by a scheduling mechanism [46]. Game-theoretic formulations capture situations in which vehicles strategically choose arrival speeds or request times, and payoffs reflect delay, collision penalties, or energy usage. In these frameworks, scheduling algorithms such as FCFS, priority rules, or graph-based schemes determine outcomes [49, 75]. However, because small timing differences can produce discrete scheduling outcomes, such systems often exhibit non-smooth payoff structures, making equilibria unstable or nonexistent [89]. Smoother priority functions or continuous reservation updates are often required to alleviate these issues.

Auction and Market-Based Priority Mechanisms

Some researchers propose allocating intersection priority using auctions, where drivers bid for faster access [90, 89]. While theoretically yielding efficient allocations, these systems introduce practical concerns such as fairness and political issues regarding monetary transfers, the need for incentive-compatible rules to prevent strategic overbidding, and demanding communication requirements for real-time operation. Although tradable-credit variants reduce the monetary burden, they still require significant accounting infrastructure [26]. As a result, such

systems remain largely theoretical.

Cooperative and Coalition-Based Behavior

Autonomous vehicles may form coalitions or platoons to secure favorable intersection access or reduce delays. Cooperative models formalize how platoons can maximize throughput by sharing benefits and internal costs [91, 92]. Coalition formation proves particularly advantageous when vehicles share intended movements, benefit from synchronized entry, or face scheduling policies that explicitly favor grouped reservations. However, these coalitions also impose externalities, potentially worsening conditions for non-cooperating vehicles or conflicting approaches.

Information Asymmetry and Signaling

Intersections often involve incomplete or asymmetric information regarding vehicle intentions, such as specific turning movements or acceleration limits. Signaling-game analyses examine how vehicles communicate or misrepresent their intentions and how control mechanisms can mitigate manipulation [32, 93]. Key insights from this literature highlight that turn signals or V2I transmissions function as either costly or cheap signals, depending on the enforcement context. While misreporting can undermine scheduling fairness and efficiency, screening mechanisms can successfully detect or constrain such manipulative behaviors. These models provide foundations for robust controller design under strategic uncertainty.

Mechanism Design for Intersection Scheduling

Mechanism design frameworks consider how to structure intersection rules such that self-interested drivers behave in ways that improve system-level performance.

Mechanisms may adjust waiting times, speed adjustments, priority scores, reservation release times, access probabilities, or timestamp offsets. Recent work uses non-monetary tools such as priorities, artificial tokens, or timestamp adjustments to influence behavior without direct payments [94, 49]. A critical insight is that intersection-level policies shape route-level costs, and therefore global equilibrium. The timestamp-based incentives developed later in this thesis build explicitly on this observation.

Spillback and Network-Level Externalities

Intersection delays propagate upstream via queue spillback. Strategic or priority decisions that seem local can have profound network-wide effects by blocking links or altering flow distributions [95, 96]. Studies incorporating spillback show that local priority decisions can significantly shift global equilibrium, and that non-monotonic node delays may prevent equilibrium existence. Consequently, bounding queue lengths or limiting spillback is often required to improve stability. This underscores the importance of realistic node-delay modeling and motivates the simulation-based validation performed in later chapters.

Summary

Game-theoretic models highlight that intersection behavior is intrinsically strategic: priority rules, scheduling mechanisms, and information signals all influence how drivers position themselves for service. These local interactions generate network-wide consequences, affecting route choice, delay patterns, and equilibrium structure. Insights from this literature motivate the timestamp-based incentive mechanism developed in this thesis, which uses controlled, path-dependent delays at intersections to influence equilibrium flows toward more efficient configurations.

2.8 Mechanism Design Perspective and the Bridge to Timestamp-Based Intersection Incentives

The preceding sections reviewed classical results in selfish routing, equilibrium inefficiency, and congestion-management strategies. Across these diverse approaches ranging from tolling and tradable credits to information provision and autonomous intersection management (AIM) a unifying theme emerges: system performance can be improved by shaping how individual drivers make routing decisions. This perspective aligns naturally with *mechanism design*, which studies how rules of interaction can be structured to produce socially desirable outcomes even when agents behave strategically according to their own objectives [97, 98]. In traffic networks, the misalignment between user equilibrium (UE) and system optimality motivates the design of mechanisms that modify drivers’ perceived costs so that self-interested choices yield efficient flows.

Mechanism Design Needs in Routing

Modern routing environments are inherently strategic, characterized by a feedback loop where drivers adapt to real-time information from navigation apps, V2I systems, and connected autonomous vehicle (CAV) platforms, while intersection controllers adapt to the resulting traffic patterns. Because these systems influence one another, intersection control cannot be designed in isolation; instead, interventions must anticipate and shape driver responses in a manner consistent with mechanism design principles. From a planner’s perspective, an effective mechanism must adjust perceived travel costs in a targeted and predictable way while operating within strict safety and capacity constraints. Furthermore, such systems should require minimal new infrastructure and induce network-wide improvements rather than merely localized gains. These objectives extend beyond classical signal optimization, which typically minimizes delay locally. In this framework, delay is used *strategically* as a tool for influencing global routing

behavior.

Limits of Monetary and Token-Based Mechanisms

Most mechanism-design approaches in transportation focus on monetary tolls or credit systems. Despite strong theoretical foundations, these approaches face notable practical challenges. For instance, tolls are often politically sensitive, require extensive billing infrastructure, and raise significant equity concerns [21, 22]. Similarly, tradable credit schemes necessitate centralized accounting and complex market platforms to function effectively [26, 25]. Auction-based priority systems face their own hurdles, as they require real-time bidding mechanisms that introduce substantial complexity and fairness issues [90, 99]. These limitations motivate incentive mechanisms that avoid monetary transactions entirely while still influencing routing decisions effectively.

Why Intersection-Level, Non-Monetary Incentives Are Attractive

Intersections naturally impose delay, making them an ideal medium for *non-monetary* incentives. Unlike tolls, delay is inherent to the physical system and universally understood by drivers. Connected and autonomous intersection controllers, especially reservation-based or virtual-queue systems can manipulate delay with high precision through mechanisms such as scheduling order, reservation timing, controlled priority, or virtual timestamp adjustments. By linking such adjustments to the path chosen by a vehicle, the intersection effectively modifies path costs. This yields a powerful idea: *intersection delay can serve as a non-monetary toll* that influences route selection while maintaining operational feasibility.

Intersection Control as a Mechanism Designer

Viewed through the lens of mechanism design, each intersection manager possesses several levers of influence: it can control waiting times, determine the service order of vehicles, and apply small path-dependent offsets to a vehicle’s effective arrival time. Classical signalized intersections provide only coarse-grained authority over these factors, but AIM systems offer much finer temporal resolution [46, 73]. This increased granularity allows intersections to implement incentive-compatible scheduling rules that intentionally modify path costs while maintaining safety.

Timestamp-Based Intersection Incentives

The timestamp-based mechanism developed in this thesis operates by introducing controlled modifications to the request times that vehicles submit when approaching an intersection. Each vehicle carries a timestamp indicating its position in the scheduling or reservation process. By advancing or delaying this timestamp based on the selected path, the system planner effectively alters the experienced delay. This mechanism is non-monetary, requires no payments or token systems, and remains fully local since intersections already manage vehicle scheduling. Because different routes lead to different timestamp adjustments, the mechanism naturally induces path-dependent delays. Even small adjustments, on the order of a few seconds, can aggregate across intersections into meaningful changes in network-level flow patterns. The approach is also well aligned with CAV technologies, whose reservation systems readily incorporate timestamp-based logic.

Analytical Tractability Within Congestion Games

A central advantage of timestamp-based incentives is that they preserve the mathematical structure of congestion games. The timestamp offset applied at node v for path p can be modeled as an additive term in the corresponding node

cost function, ensuring that full path costs remain additively separable. This compatibility allows the mechanism to leverage Beckmann-type potential functions [11], convexity properties of non-decreasing latency functions, equilibrium existence and essential uniqueness results [5], and standard bilevel optimization techniques. Such tractability is noteworthy because many AIM protocols introduce non-separable or stochastic delays that obstruct equilibrium analysis. Timestamp-based incentives, by contrast, maintain interpretability and permit rigorous mathematical treatment.

From Mechanism Design to Bilevel Optimization

Once timestamp offsets are treated as decision variables, the problem naturally assumes a bilevel structure. At the upper level, a system planner selects timestamp offsets to minimize total system cost. At the lower level, drivers perform selfish route choice, selecting the paths that minimize their perceived travel times, which include the timestamp-induced delays. This formulation mirrors Stackelberg models and inverse tolling approaches [9], but relies on delay adjustments rather than monetary charges. The bilevel model thus captures the interaction between planner and users, forming a direct conceptual bridge from mechanism design principles to the practical timestamp-based intervention developed in the subsequent chapters.

Summary

Mechanism design offers a principled framework for aligning route choices with system-wide efficiency. Intersection-level timestamp manipulation provides a realistic, non-monetary, and fine-grained method for embedding such incentives directly into traffic operations. This perspective motivates the mathematical modeling and bilevel optimization framework presented in the next chapter, where timestamp-based intersection incentives are formalized and analyzed.

Chapter 3

Strategic Control of Intersections for Efficient Traffic Routing without Tolls

In this chapter, we present the first part of our study, originally developed in [100], where we investigate how intersection-level control can be used as a non-monetary mechanism to influence route choices of selfish drivers. Building on the high-level motivation in Chapter 1, we focus on a simple but analytically rich setting: Pigou’s two-route network. Despite its simplicity, Pigou’s example is known to attain the worst-case Price of Anarchy (PoA) for Pigou-type networks, and is therefore a canonical stress-test for routing mechanisms [4].

Our goal is to understand whether suitably designed *prioritization* at a single intersection can reduce the inefficiency of user equilibrium (UE) routing in this worst-case topology, without using tolls. To this end, we compare a baseline First-Come-First-Serve (FCFS) policy with a *Strategic Priority-Based Scheduling (SPBS)* scheme that assigns different intersection priorities depending on the chosen route.

We first review the classical Pigou model and formal definitions of UE, social

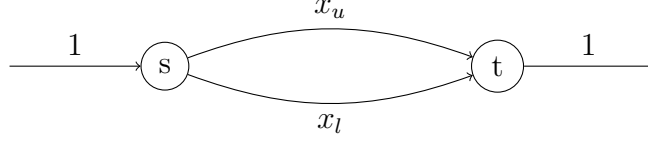


Figure 3.1: Pigou's example.

optimum, and PoA. We then introduce a simple node-delay model for FCFS and SPBS, derive closed-form expressions for UE and socially optimal flows under SPBS, and finally validate the modeling assumptions via microscopic simulations in the SUMO traffic simulator.

3.1 Preliminaries: Routing on Pigou's Network

Pigou's network consists of two parallel routes connecting a single origin s to a destination t . A unit mass of nonatomic traffic must travel from s to t . Let x_u and x_l denote the fractions of flow using the upper and lower routes, respectively:

$$x_u + x_l = 1, \quad x_u, x_l \in [0, 1]. \quad (3.1)$$

We adopt the standard choice of edge (link) cost functions:

$$c_u(x_u) := 1, \quad c_l(x_l) := x_l^n, \quad (3.2)$$

for some integer $n \in \mathbb{N}$. The upper route has a constant travel time equal to 1, while the lower route exhibits polynomial congestion.

Given a flow $\mathbf{x} = (x_u, x_l)$, the total *edge cost* experienced by all users is

$$C(\mathbf{x}) = (1 - x_l) \cdot c_u(x_u) + x_l \cdot c_l(x_l) = 1 - x_l + x_l^{n+1}. \quad (3.3)$$

User Equilibrium and Social Optimum

We recall the usual definitions for nonatomic routing games.

Definition 3.1 (User Equilibrium (UE)). Let R denote the set of routes, and let x_r and $c_r(x_r)$ be the flow and cost on route $r \in R$, respectively. A flow $\mathbf{x} = \{x_r\}_{r \in R}$ is a *user equilibrium* if, for every route r with $x_r > 0$,

$$c_r(x_r) \leq c_{r'}(x_{r'}) \quad \text{for all } r' \in R. \quad (3.4)$$

In words, no user can strictly reduce their cost by unilaterally switching routes.

In Pigou's network, this means that at UE either only one route is used (and has minimal cost), or both routes have equal cost whenever both carry positive flow.

Definition 3.2 (Socially Optimal Routing). A flow $\mathbf{x}^*$ is *socially optimal* if it minimizes the total travel cost

$$C(\mathbf{x}) = 1 - x_l + x_l^{n+1} \quad (3.5)$$

over all $x_l \in [0, 1]$.

Differentiating (3.5) with respect to x_l gives

$$-1 + (n+1)x_l^n = 0,$$

so the socially optimal fraction on the lower route is

$$x_l^* = \left(\frac{1}{n+1} \right)^{1/n}. \quad (3.6)$$

Price of Anarchy

The inefficiency of UE relative to the social optimum is quantified by the Price of Anarchy.

Definition 3.3 (Price of Anarchy (PoA)). The Price of Anarchy is defined as

$$\text{PoA} := \frac{\text{Total cost under worst UE}}{\text{Total cost under social optimum}}. \quad (3.7)$$

For polynomial latency functions, Pigou's network attains the worst possible PoA among all networks [4]. This makes it a natural benchmark for testing whether intersection-based mechanisms can improve efficiency.

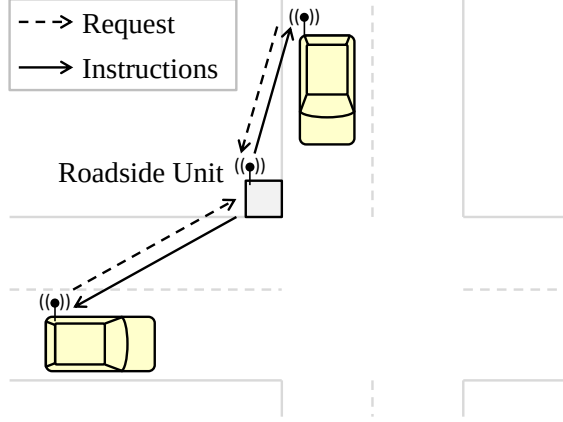


Figure 3.2: Illustrative autonomous intersection management architecture. A roadside unit receives requests from approaching vehicles and schedules their passage through the intersection.

3.2 Autonomous Intersection Management and FCFS Baseline

In Chapter 1, we outlined the concept of autonomous intersection management (AIM), where roadside units coordinate vehicles at an intersection using vehicle-to-infrastructure communication. Here we briefly revisit this idea and formalize the FCFS policy as our baseline.

AIM Architecture

In an AIM setup (Fig. 3.2), vehicles entering a predefined communication zone send a request to a roadside unit. The unit decides the order in which they are allowed to enter the conflict area, subject to safety constraints. The simplest scheduling rule is *First-Come-First-Serve (FCFS)*, where vehicles are served strictly in the order of request times.

Node Cost Model under FCFS

Under FCFS, we assume that vehicles from both routes experience the same average waiting time at the intersection. Let $w > 0$ denote this average delay. We then model the node cost as

$$d_u(x_u) = w, \quad d_l(x_l) = w, \quad (3.8)$$

for all feasible flows $\mathbf{x}$.

The total cost (edge + node) becomes

$$T(\mathbf{x}) = C(\mathbf{x}) + w = 1 - x_l + x_l^{n+1} + w. \quad (3.9)$$

The user equilibrium under FCFS coincides with the classical Pigou UE: all flow uses the lower route ($x_l^e = 1$), because for $x_l < 1$ we have $x_l^n < 1$. The socially optimal flow remains as in (3.6), and the resulting PoA matches the well-known worst-case bound for this network.

3.3 Strategic Priority-Based Scheduling (SPBS)

Motivation

While FCFS treats all vehicles identically, AIM technologies can, in principle, assign different priority levels to different classes of vehicles (e.g., public transit, emergency services). This suggests that intersection priorities can also serve as *incentives* to influence route choice. In particular, we may wish to reward drivers who choose socially beneficial routes by granting them higher intersection priority.

In the context of Pigou’s network, the FCFS equilibrium places all flow on the congestion-prone lower route, which is socially inefficient. SPBS attempts to counteract this by assigning higher priority to vehicles using the upper route, and lower priority to those on the lower route, thereby modifying the expected intersection delays in a route-dependent way.

Conceptual Description

Under SPBS, when a vehicle enters the communication zone upstream of the intersection, it submits a request that includes its route. The roadside unit assigns a dynamic score π_i to vehicle i based on a base priority level, determined by the route choice (upper vs. lower), and the time elapsed since the vehicle's request.

Vehicles with higher scores are moved toward the front of a global service queue, subject to safety constraints. A practical challenge arises because queue order must remain compatible with the physical order of vehicles on each lane. Following [66], we overcome this by temporarily boosting the scores of vehicles that would otherwise block a high-priority vehicle, ensuring that the resulting schedule is both priority-aware and physically feasible.

Average Node Costs under SPBS

To capture the effect of SPBS analytically while keeping the model simple, we adopt the following node-cost structure, which we later validate via simulation.

Assumption 3.4 (SPBS Node Cost Model). Under SPBS, the average node costs for vehicles on the upper and lower routes are given by

$$d_u(x_u) := w - p, \quad d_l(x_l) := w + p, \quad (3.10)$$

for some parameter $p \in [0, w]$. The parameter w represents a nominal intersection delay, while p captures the priority-induced difference in waiting time between the two routes. We assume the impact of $\mathbf{x}$ on the delay of external flows is negligible.

Given a flow $\mathbf{x} = (x_u, x_l)$, the total node cost under SPBS is

$$D(\mathbf{x}) = (1 - x_l)(w - p) + x_l(w + p), \quad (3.11)$$

and the total travel cost becomes

$$T(\mathbf{x}) = C(\mathbf{x}) + D(\mathbf{x}). \quad (3.12)$$

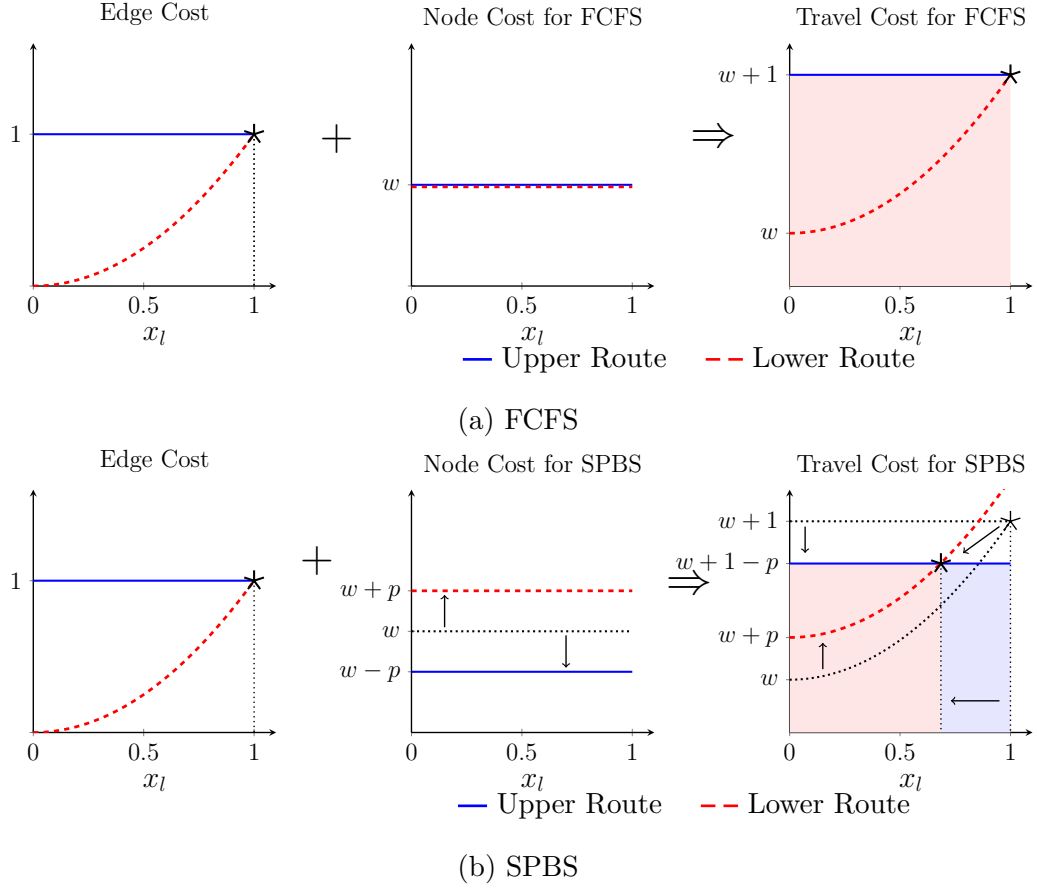


Figure 3.3: Qualitative comparison of FCFS and SPBS on Pigou's network. The top row shows how FCFS produces symmetric node delays and a UE fully on the lower route. The bottom row illustrates how SPBS introduces route-dependent node delays that shift the equilibrium toward the upper route and reduce total cost. Shaded regions represent total travel cost at equilibrium for each route.

3.4 Qualitative Cost Comparison: FCFS vs. SPBS

To illustrate how SPBS alters cost structures, we use a set of plots similar to those in the original paper. Figure 3.3 compares FCFS and SPBS in terms of edge costs, node costs, and resulting travel costs for the case $n = 2$.

Figure 3.3 highlights the key qualitative effect of SPBS: by increasing (decreasing) node delay on the lower (upper) route, the mechanism makes the upper

route relatively more attractive at equilibrium and reduces the total travel cost.

3.5 Analytical Characterization of SPBS

We now characterize the user equilibrium and socially optimal flows under the SPBS node-cost model introduced in Assumption 3.4. For clarity, we write out the total cost explicitly.

Using (3.3) and (3.11), we have

$$\begin{aligned} T(\mathbf{x}) &= 1 - x_l + x_l^{n+1} + (1 - x_l)(w - p) + x_l(w + p) \\ &= 1 + w - p - (1 - 2p)x_l + x_l^{n+1}. \end{aligned} \quad (3.13)$$

User Equilibrium under SPBS

At user equilibrium, both routes used with positive flow must have equal travel cost. The effective route costs under SPBS are

$$C_u = 1 + (w - p), \quad C_l(x_l) = x_l^n + (w + p). \quad (3.14)$$

Proposition 3.5 (UE under SPBS). *Under Assumption 3.4, the unique user equilibrium flow under SPBS is $\mathbf{x}^e = (1 - x_l^e, x_l^e)$, where*

$$x_l^e = \begin{cases} (1 - 2p)^{1/n}, & p \in [0, 1/2), \\ 0, & p \geq 1/2. \end{cases} \quad (3.15)$$

The corresponding equilibrium cost is

$$T(\mathbf{x}^e) = w + 1 - p. \quad (3.16)$$

Proof. Consider first the case where $x_l^e \in (0, 1)$ so that both routes are used. Then UE requires

$$1 + w - p = x_l^n + w + p,$$

which yields

$$x_l^n = 1 - 2p.$$

For $x_l^e \in [0, 1]$, this equation admits a solution if and only if $1 - 2p \in [0, 1]$, i.e., $p \in [0, 1/2]$. In that case,

$$x_l^e = (1 - 2p)^{1/n}.$$

If $p > 1/2$, we have $1 - 2p < 0$ and the lower route is strictly dominated by the upper route (its cost is always larger), so all flow uses the upper route and $x_l^e = 0$.

Substituting x_l^e into $C_u = 1 + w - p$ yields the equilibrium cost $T(\mathbf{x}^e) = w + 1 - p$. $\square$

Remark 3.6. For any $p > 0$, the UE cost under SPBS, $w + 1 - p$, is strictly smaller than the UE cost under FCFS, $w + 1$. Moreover, unless $p \geq 1/2$, SPBS yields an equilibrium with $x_l^e \in (0, 1)$, so both routes are used. This contrasts with the FCFS case, where at equilibrium all flow uses the lower route.

Social Optimum and PoA under SPBS

We next characterize the socially optimal flow under the SPBS model and the resulting Price of Anarchy.

Proposition 3.7 (Socially Optimal Flow under SPBS). *Under Assumption 3.4, the socially optimal flow $\mathbf{x}^* = (1 - x_l^*, x_l^*)$ under SPBS is given by*

$$x_l^* = \begin{cases} \left(\frac{1 - 2p}{n + 1} \right)^{1/n}, & p \in [0, 1/2), \\ 0, & p \geq 1/2. \end{cases} \quad (3.17)$$

The corresponding Price of Anarchy under SPBS is

$$\text{PoA} = \frac{1 + w - p}{1 + w - p - n \left(\frac{1 - 2p}{n + 1} \right)^{\frac{n+1}{n}}}. \quad (3.18)$$

Proof. From (3.13), the social cost under SPBS is

$$T(\mathbf{x}) = 1 + w - p - (1 - 2p)x_l + x_l^{n+1}.$$

To find the social optimum, we minimize this expression over $x_l \in [0, 1]$. The derivative with respect to x_l is

$$\frac{\partial T}{\partial x_l} = -(1 - 2p) + (n + 1)x_l^n.$$

Setting it to zero yields

$$x_l^n = \frac{1 - 2p}{n + 1}.$$

For $p < 1/2$, we have $1 - 2p > 0$, so $x_l^* \in (0, 1]$ is given by (3.17). For $p \geq 1/2$, the derivative is nonnegative for all $x_l \in [0, 1]$, and the minimum is attained at $x_l^* = 0$.

The PoA expression follows by evaluating $T(\mathbf{x}^e)$ at the UE flow $\mathbf{x}^e$ from Proposition 3.5, and $T(\mathbf{x}^*)$ at the SO flow $\mathbf{x}^*$ above, and then taking their ratio as in the definition of PoA. $\square$

These results demonstrate that SPBS can reduce the PoA relative to FCFS by shifting both the UE and SO flows toward the upper route and lowering the UE cost. In some regimes, the SO cost under SPBS can even be smaller than the SO cost under FCFS, emphasizing the impact of intersection priorities on overall efficiency.

3.6 Microscopic Simulation Study in SUMO

The analytical model relies on simplified, route-independent approximations of node delays parameterized by w and p . To verify that this model is consistent with realistic traffic dynamics, we conduct microscopic simulations using the SUMO traffic simulator [16].

Network and Scenario Setup

The simulated network consists of three incoming links merging into a single outgoing link. Two of the incoming links represent the upper and lower routes of

Pigou’s example, while the third corresponds to an external flow. Each incoming link is approximately 300 m in length.

Traffic demand is specified as follows. The main flow from s to t arrives at a rate of 0.15 vehicles per second, while the external flow enters the intersection at half this rate. Vehicle departures on all approaches are generated according to a Poisson process with rate 4.5, producing stochastic arrival patterns consistent with the assumptions underlying the analytical model.

For the Selective Priority-Based Scheduling (SPBS) mechanism, each vehicle i is assigned a dynamic score at time t given by

$$f_i(t, s_i) := t s_i^2, \quad (3.19)$$

where t denotes the time elapsed since vehicle i submitted its request at a point 100 m upstream of the intersection, and $s_i \in \{1, 1.5, 2\}$ represents the nominal priority level corresponding to low, medium, and high priority, respectively. Vehicles traveling on the upper route are assigned the highest priority level, those on the lower route receive the lowest priority, and vehicles belonging to the external flow are assigned medium priority.

The roadside unit (RSU) maintains a global queue of all incoming vehicles, ordered in decreasing order of the score $f_i(t, s_i)$. Vehicles are scheduled for intersection access according to this ordering, subject to standard safety constraints. The node cost is measured as the realized waiting time between the moment a vehicle enters the communication region and the moment it is granted access to the intersection.

For each configuration of the lower-route flow fraction x_l , the system is simulated over multiple independent trials, typically 100 runs of 3600 seconds each, to estimate average travel times under both first-come-first-served (FCFS) and SPBS policies.

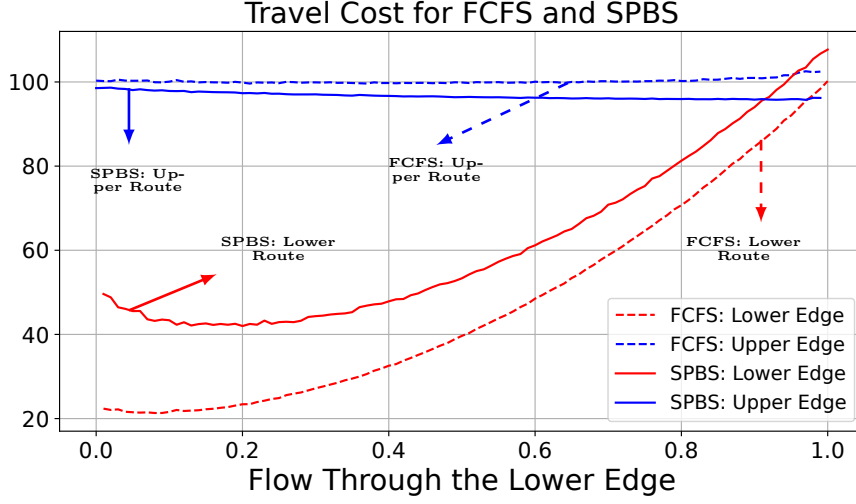


Figure 3.4: Average travel cost (edge + node) for upper and lower routes under FCFS and SPBS from SUMO simulations. The qualitative behavior is consistent with the analytical node-cost model used for SPBS.

Simulation Results

Figure 3.4 shows the average travel costs on the upper and lower routes under FCFS and SPBS, as a function of the fraction x_l of flow using the lower route. The curves closely resemble the analytical predictions in Fig. 3.3, especially in the range where $x_l \geq 0.5$.

The simulations confirm that SPBS reduces the travel cost on the favored route and increases it on the disfavored route, shifting the empirical equilibrium in the direction predicted by Proposition 3.5. Moreover, the shape of the cost curves supports the simple parametric node-delay model in Assumption 3.4, especially for practically relevant ranges of x_l and p .

3.7 Discussion and Takeaways

This chapter demonstrates that carefully designed intersection priorities can serve as effective, non-monetary incentives for mitigating inefficiencies arising from selfish routing. Even in Pigou’s example—widely regarded as a worst-case network in terms of the Price of Anarchy—appropriately chosen priority rules at intersections are sufficient to reduce equilibrium inefficiency.

The Selective Priority-Based Scheduling (SPBS) mechanism achieves this effect by introducing route-dependent delays at intersections. By granting asymmetric priority to vehicles based on their chosen routes, SPBS discourages excessive use of the more congestion-sensitive lower route and shifts the user equilibrium toward a more balanced flow distribution. As a result, the total travel cost under SPBS is consistently lower than that obtained under a first-come-first-served baseline.

Moreover, the analysis shows that a simple parametric representation of node delays—characterized by two interpretable parameters, w and p —is sufficient to capture the essential qualitative behavior of the system. Despite its simplicity, this model aligns well with the outcomes observed in microscopic simulations, indicating that the key mechanisms driving efficiency gains can be captured without resorting to overly complex delay formulations.

These findings support the broader thesis viewpoint that *intersections can be used as levers for mechanism design in traffic networks*, not only as passive bottlenecks. In the next chapter, we will generalize this insight to large networks with many intersections, by introducing a timestamp-based two-layer mechanism that coordinates path-dependent intersection incentives across the entire network.

Chapter 4

Designing Non-monetary Intersection Control Mechanisms for Efficient Selfish Routing

4.1 Overview

The previous chapter introduced the *Strategic Priority-Based Scheduling (SPBS)* mechanism developed in [100], where we demonstrated how intersection-level priority adjustments can reshape routing behavior in simple networks such as Pigou’s and verified these effects through microscopic traffic simulations. While SPBS provides valuable insight into how local intersection policies influence global traffic patterns, its modeling scope is intentionally narrow: it does not incorporate timestamp-based coordination across multiple intersections, nor does it establish a general analytical foundation for large and heterogeneous networks. [101] addresses a substantially broader question:

How can a system planner systematically influence selfish routing decisions across an entire transportation network by using non-monetary, timestamp-based interventions at autonomous intersections?

To answer this question, [101] introduces a *network-wide* mechanism in which the planner modifies the timestamps submitted to roadside units (RSUs). These modifications act as *delays* or *advancements* and indirectly influence the expected waiting time of a driver at each intersection. [101] makes the timing-based nature of intersection control *explicit* and models its influence as additive, path-dependent node costs.

Relation to Chapter 3. Chapter 3 focused exclusively on the SPBS mechanism and the analytical and experimental results obtained from [100]. Although SPBS laid the conceptual foundation for understanding how intersection-level prioritization changes driver incentives, it does not generalize to arbitrary network topologies nor does it provide a theoretical framework suitable for equilibrium analysis, potential-function formulations, or bilevel optimization.

The timestamp mechanism introduced in this chapter generalizes the intuition developed in Chapter 3 in two important ways. First, it explicitly incorporates intersection delays into the congestion-game framework by introducing node-level cost terms. Second, it enables a system planner to shape routing decisions through path-dependent incentives that can be tuned offline via optimization. As a result, Chapter 4 establishes the analytical backbone upon which the full two-layer architecture, presented later, is constructed.

Why Path-Dependent Node Costs Are Introduced. Classical congestion games represent delay primarily through link travel times and typically treat intersections implicitly. In contrast, modern autonomous intersection management enabled by vehicle-to-infrastructure communication makes the *intersection service process* explicit: vehicles submit requests, and a roadside unit schedules their access to the conflict zone subject to safety and feasibility constraints. The waiting time that results is therefore shaped not only by the total demand at the intersection, but also by how requests are ordered and timed under the intersection’s scheduling protocol.

Because the system operator can influence this ordering through small timestamp adjustments (e.g., advancing or delaying the effective request time), the

induced intersection delay can differ across vehicles that traverse the same intersection depending on the routes they choose. This creates an inherently *path-dependent* intersection effect, even though the physical bottleneck is a node.

This motivates a modeling approach in which intersection control effects are captured as additive, path-specific adjustments to a baseline node delay. Analytically, this structure integrates cleanly into a potential-based equilibrium framework and preserves tractability and key equilibrium properties. Empirically, microscopic SUMO experiments indicate that timestamp modifications largely act as vertical shifts of the measured intersection-delay curves, supporting an additive modeling interpretation. Together, these observations justify introducing path-dependent node costs while retaining desirable properties such as existence and essential uniqueness of equilibrium flows.

Purpose of Chapter 4. The goal of this chapter is to develop the complete analytical machinery underlying the proposed *two-layer incentive architecture*, in which a system planner designs intersection-level timestamp incentives while anticipating the equilibrium response of selfish drivers. Specifically, this chapter aims to:

1. model timestamp-based incentives through additive, path-specific node costs,
2. characterize the induced equilibrium behavior using a generalized potential function,
3. establish existence and essential uniqueness of equilibrium flows under these incentives,
4. formulate the planner’s decision problem as a bilevel optimization program that explicitly links incentive design to selfish routing responses.

Together, these results extend classical congestion-game theory and provide the analytical backbone for the two-layer framework, preparing the ground for the numerical and experimental validation presented in Chapter 5.

4.2 Proposed Two-Layer Intersection Control Mechanism

Modern connected and autonomous vehicles provide a unique opportunity to rethink how intersections operate and how individual travel decisions influence network-wide congestion patterns. Unlike traditional traffic signals, autonomous intersections allow roadside units (RSUs) to schedule vehicles *one by one* based on information transmitted through vehicle-to-infrastructure communication channels. This fine-grained control forms the foundation of our mechanism.

While Chapter 3 demonstrated through the Strategic Priority-Based Scheduling (SPBS) policy how local prioritization can reshape equilibrium behavior even in a simple two-road setting, the approach in that chapter remained limited to single-intersection scenarios. In contrast, the mechanism presented in this chapter generalizes these ideas to *arbitrary networks* with many intersections and multiple competing routes. This extension requires an architectural design that can influence routing choices not only at one intersection but across the entire network.

System Architecture

Our mechanism adopts a **two-layer architecture** consisting of (i) a *high-level system planner* and (ii) *local roadside units (RSUs)*. Figure 1.2 illustrates the autonomous-intersection environment in which vehicles communicate with RSUs, while Fig. 4.1 summarizes the overall structure of the proposed mechanism.

As a vehicle approaches an intersection, it transmits a request packet to the corresponding RSU. This packet usually includes a unique vehicle identifier, information about the vehicle’s intended turning movement or local routing plan, and a timestamp indicating the requested entry time into the intersection. RSUs use

these timestamps as key inputs to their local scheduling algorithms. In many commonly used policies, such as *first-come-first-served*, the relative ordering of timestamps directly determines service order, with earlier requests typically receiving earlier access. More advanced scheduling approaches including trajectory-based optimization and graph-based conflict-resolution method retain this fundamental structure: earlier requests are generally associated with shorter waiting times, even when additional safety and feasibility constraints are imposed.

High-Level Planner and Timestamp Adjustments

The key novelty of the proposed mechanism lies in how the high-level planner interacts with local intersection controllers. Rather than modifying physical infrastructure or imposing monetary tolls, the planner intervenes digitally by adjusting the timestamps contained in drivers' request packets *before* these requests are processed by the corresponding RSUs.

These timestamp adjustments act as **non-monetary incentives**. When a timestamp is delayed, the RSU processes the request as if the vehicle had arrived later, which typically results in a longer waiting time before intersection access. Conversely, when a timestamp is advanced, the RSU treats the vehicle as an earlier arrival, potentially granting earlier passage and reducing its waiting time. Through such adjustments, the planner can systematically influence the relative attractiveness of different routes. For example, if a particular corridor is overutilized at equilibrium, delaying timestamps for vehicles using that corridor encourages some drivers to divert toward alternative paths. Similarly, advancing timestamps along underutilized routes increases their appeal.

This design offers several strategic advantages. It avoids the use of monetary tolls and thus sidesteps equity and political concerns associated with pricing mechanisms. It relies purely on digital interventions without altering road geometry or physical infrastructure. Moreover, it is compatible with a wide range of RSU scheduling policies because it affects only the ordering of requests rather than the scheduling algorithm itself. Finally, the architecture scales naturally to

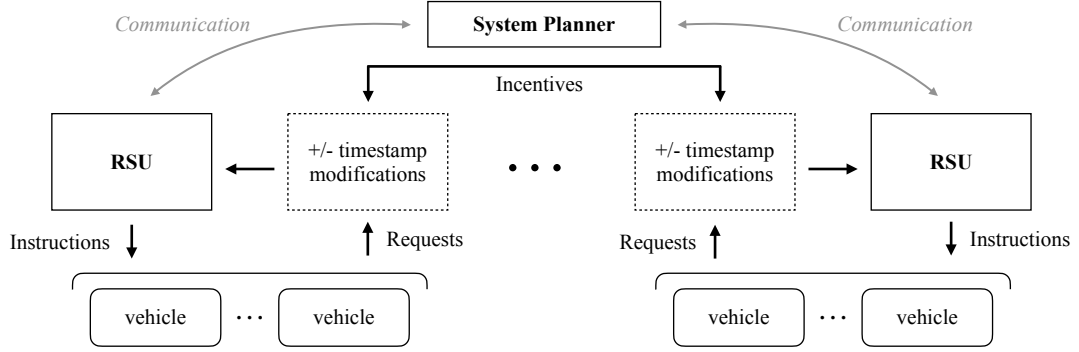


Figure 4.1: Two-layer mechanism: vehicles submit requests with timestamps, the system planner applies timestamp adjustments to generate non-monetary incentives, and RSUs schedule intersection usage based on the modified timestamps.

large networks, as RSUs operate locally while the planner coordinates incentives at the network level.

Operational Flow

In operation, vehicles approaching intersections transmit request packets containing their identifiers, timestamps, intended movements, and possibly route information. The high-level planner examines the intended paths and applies timestamp advancements or delays according to the current network-wide incentive policy. These modified timestamps are then forwarded to the local RSUs, which execute their scheduling algorithms using the adjusted requests and assign crossing slots accordingly.

To ensure safety and operational consistency, timestamp adjustments must respect practical constraints. In particular, they must not reverse the relative order of vehicles traveling in the same lane or violate minimum headway requirements needed to prevent collisions. These constraints guarantee that the mechanism influences only the *relative desirability* of routes while remaining compatible with real-world intersection operations.

Why Path-Dependent Intersection Costs?

Timestamp adjustments influence how RSUs prioritize vehicles at intersections. From the perspective of a routing game, this implies that a route accumulates additional delay at each intersection it traverses, depending on how its vehicles' timestamps are modified. Importantly, this delay depends on the chosen path rather than on the identity of individual drivers, creating a systematic and predictable incentive structure.

In essence, timestamp modifications alter the experienced intersection delay in a path-dependent manner. This observation motivates the introduction of **path-dependent node costs**, which capture how routing decisions interact with intersection scheduling outcomes. These costs generalize the ideas introduced in Chapter 3: instead of prioritizing routes at a single intersection, the planner now coordinates timestamp adjustments across multiple intersections to influence routing behavior at the network level.

This discussion establishes the conceptual foundation for the analytical framework developed in the next section. In Section 4.3, we formally model routing games with path-dependent node costs, characterize the resulting equilibrium behavior, and study how timestamp-based incentives can be optimized across the entire network.

4.3 Problem Formulation

We consider a traffic network composed of multiple autonomous intersections, each equipped with a roadside unit (RSU). Vehicles approaching an intersection transmit request packets that include identification information and timestamps, allowing the system planner to track the time at which each vehicle requests access to the intersection. In settings where vehicles optionally transmit their routing plans, the planner can also infer a vehicle's full trajectory from the origin terminal to the relevant intersection (see Fig. 1.2). RSUs process these requests

using their local scheduling policies—such as first-come-first-served [46, 47] or more sophisticated planning-based controllers [102, 103, 104]. In all cases, earlier timestamps generally translate to earlier service and shorter intersection-related delays.

We introduce an additional decision-making layer across the network, operated by a central system planner. This planner modifies the timestamps of vehicles prior to RSU scheduling. Timestamp advancements serve as positive incentives, while timestamp delays act as negative incentives. As illustrated in Fig. 4.1, RSUs perform their scheduling based on these adjusted timestamps. The goal is to alter the relative attractiveness of different routes, thereby influencing routing decisions toward a more efficient distribution of traffic. In the remainder of this section, we formalize timestamp offsets as *path-dependent node costs*.

4.3.1 Analytical Framework

We model the traffic network as a directed graph $G = (V, E)$, where V is the set of intersections and E the set of connecting roads. For ease of notation, we consider a single origin–destination pair (s, t) with a unit mass of traffic. Let $\mathcal{P}$ denote the finite set of all simple s – t paths. A traffic flow is represented as a vector $f = (f_p)_{p \in \mathcal{P}}$, where f_p is the fraction of users selecting path p . We assume fixed (inelastic) demand, so feasible flows satisfy

$$\mathcal{F} := \left\{ f = (f_p)_{p \in \mathcal{P}} : \sum_{p \in \mathcal{P}} f_p = 1, f_p \geq 0 \forall p \right\}. \quad (4.1)$$

For any edge $e \in E$ and node $v \in V$, the induced edge and node flows are

$$f_e := \sum_{p \in \mathcal{P}: e \in p} f_p, \quad f_v := \sum_{p \in \mathcal{P}: v \in p} f_p. \quad (4.2)$$

Each edge e is associated with a congestion-dependent cost function $c_e : [0, 1] \rightarrow [0, \infty)$, corresponding to the mean travel time on that road segment. Distinct from classical routing games, each intersection v also incurs a *path-dependent*

node cost $c_v^p(f_v, u_v^p)$, which incorporates a *base delay* $c_v(f_v)$ determined by congestion and RSU scheduling behavior, and a *timestamp-induced offset* $u_v^p \in [-\underline{\tau}, \bar{\tau}]$ chosen by the system planner.

The node cost therefore takes the form

$$c_v^p(f_v, u_v^p) := c_v(f_v) + u_v^p, \quad (4.3)$$

where $c_v : [0, 1] \rightarrow [\underline{\tau}, \infty)$ denotes the congestion baseline and the bounded interval for u_v^p limits how aggressively timestamps can be advanced or delayed.¹

Assumption 4.1. All edge cost functions $c_e(\cdot)$ and node cost functions $c_v^p(\cdot)$ are nonnegative, continuous, and nondecreasing in their flow arguments.

Remark 4.2 (Average Behavior). Timestamp modifications occasionally alter the order of vehicles arriving in the same lane. These local inconsistencies are resolved operationally by RSUs (e.g., by temporary reordering or lane-level heuristics). Because vehicle arrivals and route choices are stochastic, individual anomalies average out, and expression (4.3) captures the *expected* intersection delay experienced by flow f_p .

The cost of a path p under flow f and incentive profile $u = (u_v^p)$ is

$$c_p(f, u) := \sum_{e \in p} c_e(f_e) + \sum_{v \in p} c_v^p(f_v, u_v^p). \quad (4.4)$$

Drivers act selfishly and select minimum-cost paths. This leads to the standard notion of Wardrop equilibrium.

Definition 4.3 (Wardrop Equilibrium). For a fixed incentive profile u , a feasible flow $f \in \mathcal{F}$ is a Wardrop equilibrium if

$$c_p(f, u) \leq c_{p'}(f, u) \quad \forall p, p' \in \mathcal{P} \text{ with } f_p > 0. \quad (4.5)$$

Remark 4.4 (Equilibrium Interpretation). Under fixed timestamp offsets u , agents repeatedly adjust their route choices until stabilizing on paths of minimal cost.

¹The condition $\underline{\tau} \geq 0$ guarantees nonnegativity of all node costs.

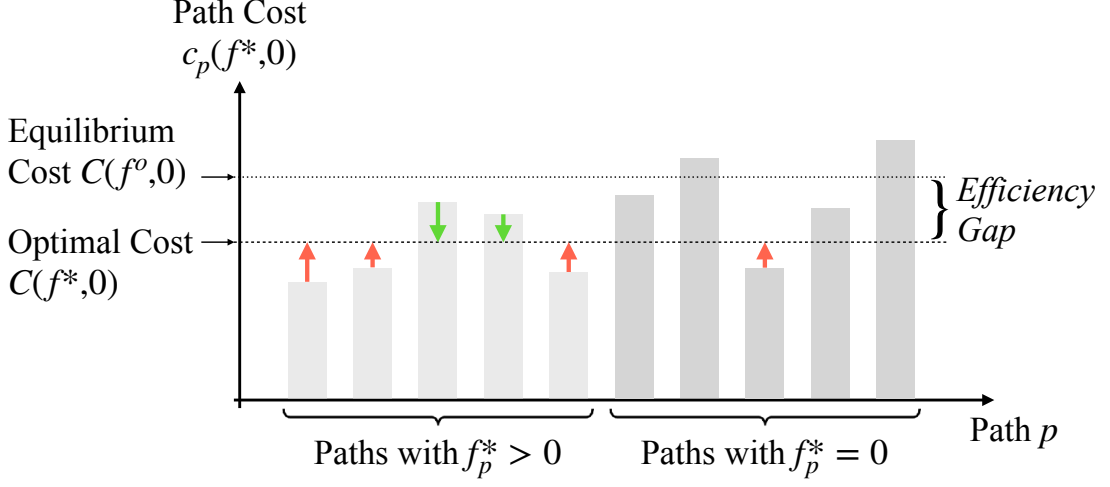


Figure 4.2: Illustration of path costs under the optimal flow f^* in the absence of incentives. Timestamp-based adjustments can raise or lower path costs to ensure that f^* satisfies the Wardrop equilibrium inequalities.

Wardrop equilibrium captures this long-run distribution. It does not require real-time optimization by drivers; it describes the emergent flow pattern when incentives remain fixed and drivers learn or perceive the effective path costs.

The social cost under incentives u is defined as

$$C(f, u) := \sum_{p \in \mathcal{P}} f_p c_p(f, u). \quad (4.6)$$

Substituting (4.4) into (4.6) and regrouping terms yields:

$$C(f, u) = \sum_{e \in E} c_e(f_e) f_e + \sum_{v \in V} c_v(f_v) f_v + \sum_{p \in \mathcal{P}} f_p u_p,$$

where $u_p := \sum_{v \in p} u_v^p$ is the total timestamp offset accumulated along path p . The first two terms represent congestion-driven costs without incentives:

$$C^o(f) := \sum_{e \in E} c_e(f_e) f_e + \sum_{v \in V} c_v(f_v) f_v. \quad (4.7)$$

Without incentives ($u_v^p = 0$), optimal flows need not coincide with equilibrium flows. As shown in Fig. 4.2, some paths used in the socially optimal flow f^* may

have strictly higher cost than unused paths, producing an efficiency gap. Ideally, incentives u should be selected so that all paths used in f^* have identical (and minimum) cost under u , and all unused paths have weakly higher cost, thereby rendering the optimum flow f^* an equilibrium. However, because timestamp modifications are bounded and may be subject to structural constraints (e.g., indistinguishability of partially overlapping routes), such an ideal adjustment may not always be feasible.

Remark 4.5 (Practical Constraints). The feasible incentive set $\mathcal{U}$ may incorporate a variety of operational restrictions arising from real-world deployment considerations. In particular, timestamp adjustments may be required to remain within prescribed bounds, such as fixed lower and upper limits on admissible advancements or delays. In addition, incentives may not be fully path-specific when routes share common prefixes, since vehicles cannot be distinguished before they reach diverging intersections. Finally, limitations imposed by local RSU scheduling policies may further restrict the structure or magnitude of admissible incentives.

If vehicles truthfully report their intended routes, then incentives can be designed to be fully path-dependent. Otherwise, the planner may only be able to differentiate between paths based on their observable prefixes.

Let $\mathcal{U} \subset \prod_{v,p} [-\underline{\tau}, \bar{\tau}]$ denote the set of all implementable incentive profiles. We can now state the optimal incentive design problem.

Problem 4.6 (Optimal Incentive Design). *Determine*

$$\min_{u \in \mathcal{U}} \max_{\substack{f \in \mathcal{F} \\ f \text{ is an equilibrium under } u}} \{C^o(f) + f \cdot u\}, \quad (4.8)$$

where $f \cdot u := \sum_p f_p u_p$ denotes the flow-weighted incentive cost.

If multiple equilibrium flows exist for a given u , Problem 4.6 evaluates the *worst-case* equilibrium cost, ensuring robustness. Later, we show that under mild conditions, equilibrium flows are essentially unique, simplifying the analysis.

4.4 Equilibrium Characterization

Our timestamp-based mechanism introduces path-dependent node costs that modify how individual drivers perceive the cost of any given route. Because drivers behave selfishly, the resulting traffic pattern is determined not directly by the system planner, but by the equilibrium induced by the incentives. Therefore, before we design optimal incentives, we must characterize the resulting equilibrium flows under a fixed incentive profile. This section establishes such a characterization by constructing a suitable potential function and proving foundational properties such as existence and essential uniqueness.

Classical routing models typically assign costs only to edges, with nodes serving as structural connectors without associated travel times, e.g., [4]. In contrast, our timestamp-based mechanism introduces *additive, path-specific intersection delays*. These additional costs break the conventional structure in which all users experience the same cost on each resource: users traversing the same intersection may now experience different costs depending on their chosen path. The following remark clarifies how our formulation relates to existing models.

Remark 4.7 (Different Intersection Costs). Congestion games with heterogeneous or player-specific costs allow different agents to incur different costs on the same resource [105, 106]. Such models capture rich behavioral patterns but may lose desirable equilibrium properties such as existence, uniqueness, or a potential structure. Variants with additive or multiplicative offsets have been studied in [107, 108], where carefully structured offsets preserve potential functions. Our model also exhibits path-dependent costs at intersections; however, the costs are *not* driver-specific. Any driver choosing the same path incurs the same node cost. Thus, our model remains aligned with the class of unweighted congestion games, preserving the desirable equilibrium properties needed for tractable analysis.

A second challenge arises from the fact that path-specific node costs depend on the *total* flow through an intersection, not on the flow of one path alone. This prevents us from collapsing node and edge costs into a single set of edge costs in the traditional sense. To address this issue, we show that the network can be

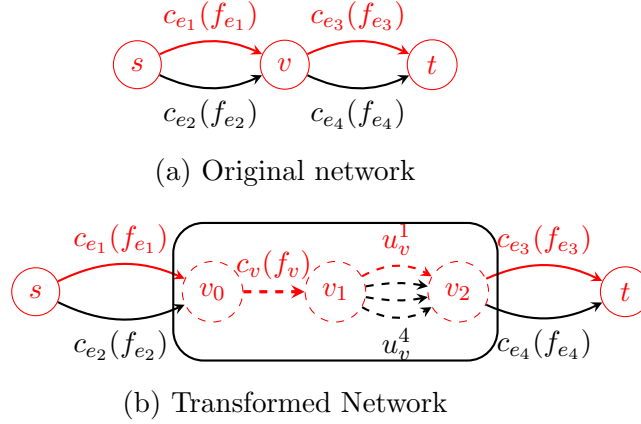


Figure 4.3: Comparison of the original (a) three-node/four-edge network and (b) the transformed version with node splitting. The path $p_1 = se_1ve_3t$ and its counterpart in the transformed network are highlighted in red.

transformed into an equivalent classical congestion network using a node-splitting construction. The following illustrative example summarizes the idea.

Example 4.8 (Node-Splitting Construction). Consider the network in Fig. 4.3a with node v and four paths $p_1, \dots, p_4$. Each path experiences a distinct node cost at v . In Fig. 4.3b, node v is replaced by two components: a baseline node-cost arc capturing $c_v(f_v)$, and a collection of path-specific arcs capturing the additive offsets u_v^p . This converts the problem into a standard congestion network, allowing application of classical routing-game tools.

This transformation ensures that path-dependent node costs can be handled within a classical potential-game framework. We now proceed to formal equilibrium characterization.

Based on the hyper-edge transformation, we define the potential function

$$\Phi(f, u) := \sum_{e \in E} \int_0^{f_e} c_e(z) dz + \sum_{v \in V} \int_0^{f_v} c_v(y) dy + f \cdot u, \quad (4.9)$$

which extends Beckmann's classical formulation in which its minimizers coincide with equilibrium flows[11] to account for intersection delays and timestamp-based incentives.

Proposition 4.9 (Potential Function). *Consider a routing game additive path-specific costs. Under Assumption 4.1, and for a fixed incentive profile $u \in \mathcal{U}$, a feasible flow f is a Wardrop equilibrium if and only if*

$$f \in \arg \min_{\tilde{f} \in \mathcal{F}} \Phi(\tilde{f}, u). \quad (4.10)$$

The proof (Appendix A) extends the classical characterization of Wardrop equilibrium [109] to accommodate the additive path-specific costs.

Our next result establishes existence and uniqueness properties of equilibrium flows.

Theorem 4.10 (Existence and Essential Uniqueness). *Consider a routing game with additive path-specific costs under a given exogenous incentive profile u . Under Assumption 4.1:*

- **Existence:** *An equilibrium flow always exists*
- **Essential Uniqueness:** *For any two equilibrium flows f and $\bar{f}$, the social cost is identical, i.e.,*

$$C(f, u) = C(\bar{f}, u).$$

The proof (Appendix B) follows from an extension of standard results in classical routing games. The essential uniqueness of equilibrium flows implies that from the worstcase to the best-case, all equilibrium improvements lead to the same problem formulation in Problem 4.6.

4.5 Bilevel Optimization Formulation

Proposition 4.9 shows that in the absence of timestamp-based incentives, drivers' equilibrium behavior is governed by the classical potential

$$\Phi^o(f) = \sum_{e \in E} \int_0^{f_e} c_e(x) dx + \sum_{v \in V} \int_0^{f_v} c_v(y) dy. \quad (4.11)$$

Thus, the unconstrained equilibrium flows correspond to the minimizers of $\Phi^o(f)$ over the feasible set $\mathcal{F}$. When incentives u are introduced, drivers perceive path costs shifted by the additive terms u_p , and the new equilibrium is induced by the modified potential $\Phi(f, u) = \Phi^o(f) + f \cdot u$. By Proposition 4.9, all minimizers of $\Phi(f, u)$ yield the same social cost.

This observation allows us to rewrite Problem 4.6 as the following bilevel optimization problem. The upper level determines the incentive vector $u \in \mathcal{U}$, while the lower level describes the drivers' collective response through the minimization of the modified potential.

Problem 4.11 (Bilevel Optimization Problem). *Solve*

$$\min_{u \in \mathcal{U}} \{ C^o(f(u)) + f(u) \cdot u \}, \quad (4.12)$$

subject to the lower-level equilibrium condition

$$f(u) \in \arg \min_{\tilde{f} \in \mathcal{F}} \{ \Phi^o(\tilde{f}) + \tilde{f} \cdot u \}. \quad (4.13)$$

In this formulation, the lower-level problem characterizes the decisions of a large population of selfish drivers. For a fixed incentive profile u , drivers select routes that minimize their perceived travel costs. Proposition 4.9 ensures that this behavior is equivalent to minimizing the potential $\Phi^o(f) + f \cdot u$. Because all equilibrium flows under a given incentive vector u yield the same social cost, the upper-level objective in Problem 4.11 is well-defined.

The upper-level planner, representing a transportation authority or automated system controller, aims to select incentives u to minimize the resulting system-wide travel cost. However, the planner must account for the fact that f is not an independent variable but it is the equilibrium outcome $f(u)$ of the lower level problem. This structure naturally yields a two-stage *Stackelberg game*: the planner acts as the leader, announcing incentives first, and the drivers act as followers, collectively adjusting their behavior to minimize individual perceived costs.

The upper-level objective

$$C^o(f) + f \cdot u$$

captures the fundamental trade-off induced by timestamp modifications. While incentives can shift equilibrium flows toward more efficient traffic patterns, they also directly contribute to perceived costs along each path. Since both the baseline routing costs and incentive offsets appear in the social cost, the optimal design must balance improving the equilibrium routing pattern with ensuring that incentives do not inflate total travel cost excessively.

Because $f(u)$ is implicitly defined by the lower-level equilibrium problem, Problem 4.11 is a nontrivial optimization problem. Moreover, the feasible set $\mathcal{U}$ may include operational constraints, such as bound limits on timestamp advances or delays and feasibility conditions arising from RSU scheduling.

4.6 Illustrative Examples

This section provides a detailed investigation of how our proposed timestamp modification mechanism improves routing efficiency in networks with selfish drivers. First, we validate the analytical framework using a canonical example, the Braess network, which is well known for demonstrating how individually rational routing choices may degrade system performance. We further validate the analytical model of additive path-dependent costs induced by timestamp modification and examine its effectiveness in high-fidelity traffic simulations of the Sioux Falls transportation network.

4.6.1 Braess Network

The first example focuses on the classical Braess network, depicted in Fig. 4.4, which consists of two parallel routes from s to t , connected by an additional zero-cost link. The counterintuitive feature of this network is that the addition of a seemingly beneficial shortcut (the central link) can induce an equilibrium that performs strictly worse than the equilibrium in the network without that link. This paradox exemplifies how selfish routing may lead to inefficient outcomes and

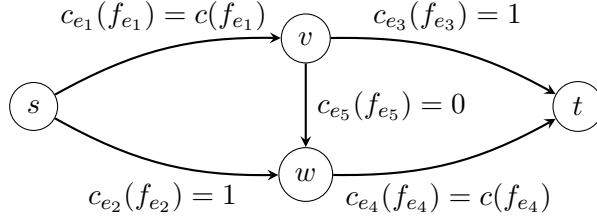


Figure 4.4: A Braess network.

Table 4.1: Equilibrium flows and social costs in the Braess network under two cost models.

Cost Model	Quadratic		Quartic	
	Flow	Cost	Flow	Cost
Equilibrium without Incentives	(0.41, 0.17, 0.41)	2	(0.34, 0.31, 0.34)	2
Optimal without Incentives	(0.5, 0, 0.5)	1.87	(0.5, 0, 0.5)	1.69
Equilibrium with Incentives	(0.5, 0, 0.5)	1.87	(0.5, 0, 0.5)	1.69

provides a natural testbed for evaluating the potential improvements offered by timestamp-based incentives. The three s - t paths are

$$\mathcal{P} = \{p_1 = se_1ve_3t, p_2 = se_1ve_5we_4t, p_3 = se_2we_4t\}.$$

To evaluate how timestamp incentives influence the resulting equilibria, we consider two families of latency functions for edges e_1 and e_4 :

- Quadratic model: $c(f) = f - 0.5f^2$
- Quartic model: $c(f) = 0.5f - 0.5f^2 - f^3 + 2f^4$

Both models satisfy Assumption 4.1 and capture increasingly nonlinear congestion effects. Node delays at v and w are set as $c_v(f_v) = f_v$ and $c_w(f_w) = f_w$. To illustrate the role of timestamp offsets, we restrict incentives to non-negative values:

$$u_v^p, u_w^p \in [0, 0.2],$$

Table 4.2: Equilibria and social costs in the Braess network under two cost models.

Cost Model	Quadratic		Quartic	
	Flow	Cost	Flow	Cost
Equilibrium without Incentives	(0.41, 0.17, 0.41)	2	(0.34, 0.31, 0.34)	2
Optimal without Incentives	(0.5, 0, 0.5)	1.87	(0.5, 0, 0.5)	1.69
Equilibrium with Incentives	(0.5, 0, 0.5)	1.87	(0.5, 0, 0.5)	1.69

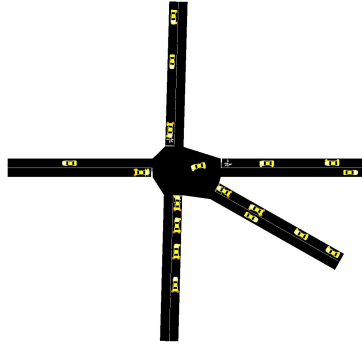
so that incentives correspond exclusively to *delays*, i.e., timestamp increases. This modeling choice captures restrictive operational settings where RSUs may not be able to advance a timestamp (e.g., due to safety or platooning constraints), but are allowed to postpone a vehicle.

We solve the bilevel problem in (4.12) using several global optimization methods, including simplicial homology global optimization algorithm (SHGO) [110], differential evolution (DE) [111], and dual annealing (DA) [112]. All algorithms converged to the optimal incentive structure

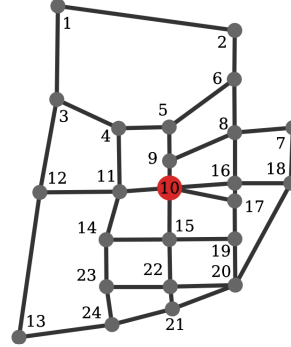
$$(u_v^{p_1}, u_v^{p_2}, u_w^{p_2}, u_w^{p_3}) = (0, 0.2, 0.2, 0),$$

which delays only path p_2 . This aligns with intuition: in the Braess network, the inefficiency arises precisely because the zero-cost link promotes overuse of the middle path p_2 ; delaying this path shifts the equilibrium back toward the more balanced socially optimal distribution. Table 4.2 summarizes the key outcomes.

The results reveal that the timestamp modifications successfully steer traffic to the socially optimal pattern for both latency models. In particular, the inefficient usage of the central link is eliminated. The fact that relatively small incentive bounds suffice to recover the optimal flow underscores the effectiveness of the



(a) SUMO intersection model for Node 10 (25 m control zone).



(b) Sioux Falls network topology (24 nodes, 76 links).

Figure 4.5: Microscopic–macroscopic linkage: a representative intersection simulated in SUMO (left) and the full Sioux Falls network (right).

mechanism in networks prone to self-induced congestion.

4.6.2 Experimental Analyses on the Sioux Falls Network

We conduct extensive experimental analyses to bridge the analytical framework developed in this chapter with realistic microscopic traffic behavior. The goal is to assess whether the proposed timestamp-based incentive mechanism yields meaningful efficiency gains when deployed on a realistic urban network whose intersection delays are empirically calibrated.

We employ the Sioux Falls transportation network, a widely used benchmark in transportation engineering and algorithmic game theory consisting of 24 nodes, 76 directed links, and 552 OD pairs [113]. The network provides a medium-scale urban topology with empirically calibrated BPR link-cost functions but does not include intersection-level delays. To incorporate realistic node costs, we construct microscopic simulations for each intersection using the Simulation of Urban Mobility (SUMO) platform [16].

Each of the 24 intersections is modeled as an independent SUMO scenario that replicates its geometry, lane configuration, and control policy. Vehicle arrivals follow a Poisson process, introducing stochastic queueing patterns representative of

real traffic conditions. For each simulation run, we record vehicle travel times from a point 25 m upstream of the stop line (corresponding to the RSU control zone) until the vehicle fully exits the intersection. This measurement captures both queueing delay and traversal time. Averaging across randomized runs produces smooth and repeatable estimates of node traversal delay as a function of incoming flow.

4.6.2.0.1 Base Node-Cost Fitting. The measured intersection delays are fitted to quartic polynomials of the form

$$c_v(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0, \quad (4.14)$$

for each node. This choice ensures consistency with the classical BPR edge-cost model, which also exhibits fourth-order dependence on flow, enabling a unified treatment of link and node congestion. Across all intersections, the quartic models achieve high accuracy, with mean absolute percentage errors typically between 1.5% and 2.5%. Table C.1 lists all coefficients and mean absolute percentage errors (MAPE) for all nodes.

4.6.2.0.2 Validation of the Additive Incentive Model. To validate the modeling assumption that timestamp-based incentives act as additive shifts in node costs, we conduct controlled SUMO experiments on a representative intersection (Node 10). For one designated path, vehicle request timestamps are modified by ± 10 seconds, representing timestamp advancement and delay, respectively. These adjustments are applied subject to practical constraints: timestamp modifications are bounded so as not to violate vehicle ordering or minimum headway requirements within the same lane.

The resulting average traversal-time curves, shown in Fig. 4.6, exhibit approximately parallel translations relative to the baseline case without incentives. This confirms that timestamp offsets primarily induce vertical shifts in the node-cost function without altering its dependence on flow, empirically validating the additive cost structure assumed in the analytical model.

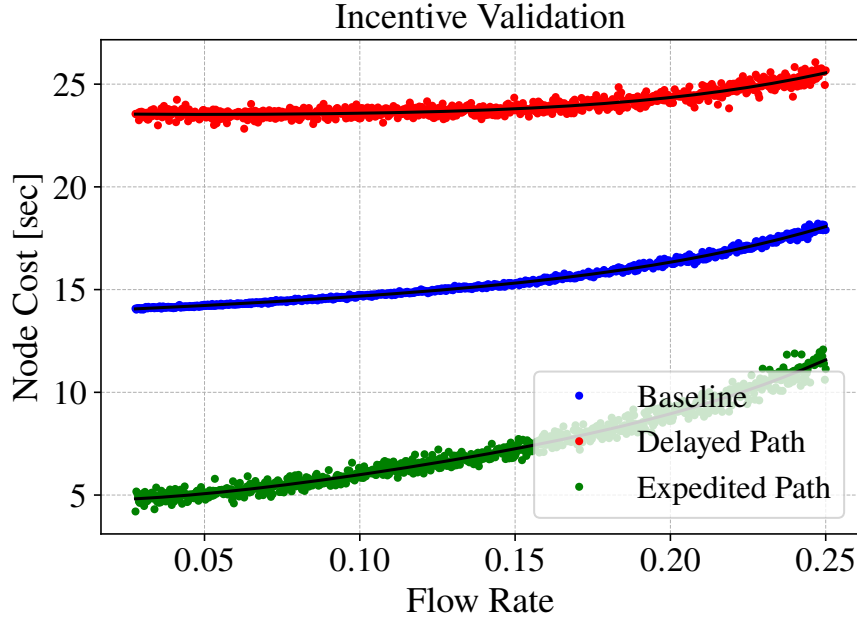


Figure 4.6: A plot of average travel time (second) versus flow rate (vehicles per second) within the control zone and the flow rate for the simulated Node 10 using the FCFS protocol under no, positive and negative timestamp modifications for one of the paths.

4.6.2.0.3 Network-Wide Optimization. The fitted node-cost functions are combined with BPR link costs to form a unified macroscopic congestion model. User equilibria are computed using the Frank–Wolfe algorithm, while the upper-level incentive design problem is solved using the Simultaneous Perturbation Stochastic Approximation (SPSA) method [114]. We consider two incentive regimes to assess both theoretical and operational trade-offs: a high-intervention case with timestamp offsets bounded by $[0, 2]$ minutes, and a low-intervention case with offsets bounded by $[0, 0.5]$ minutes (30 seconds). All simulation and algorithmic parameters are detailed in Appendix C.

Fig. 4.7 illustrates the convergence of social cost over SPSA iterations. In both cases, the incentivized equilibria steadily approach the system-optimal benchmark, while the baseline user equilibrium remains substantially higher. Quantitatively, the baseline UE cost is 8.04×10^6 minutes, compared to a system-optimal value of 7.75×10^6 minutes. The high-intervention regime achieves a final cost of 7.84×10^6 minutes, closing approximately 68.2% of the optimality gap, while the

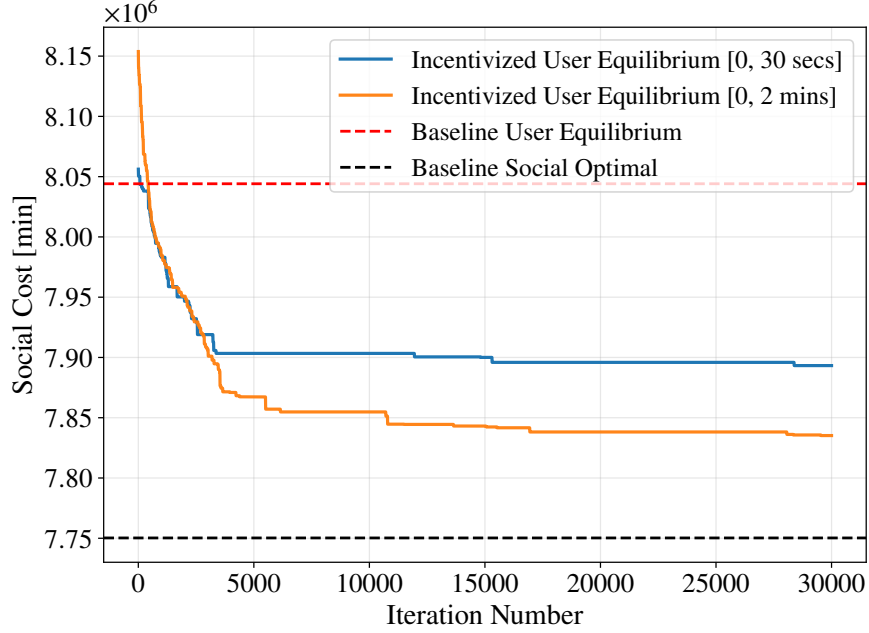


Figure 4.7: Convergence of social cost under SPSA optimization for different incentive bounds.

low-intervention regime achieves 7.93×10^6 minutes, closing 40.1% of the gap.

Fig. 4.8 provides a visual comparison of baseline and incentivized flow patterns in the Sioux Falls network under the $[0, 2]$ -min incentive regime. The baseline user-equilibrium flow distribution, shown in Fig. 4.8a, reveals a pronounced concentration of traffic along a small number of central corridors, indicating significant congestion and uneven utilization of network capacity. In contrast, Fig. 4.8b depicts the changes induced by timestamp-based incentives, where red links correspond to increased flow and blue links to reduced flow relative to the baseline. These differences highlight how the proposed incentives systematically relieve pressure on over-saturated central routes by redirecting a portion of the demand toward less congested peripheral paths. As a result, traffic is distributed more evenly across the network, explaining the observed reductions in total travel time. Together with the convergence and cost results, these visualizations confirm that timestamp adjustments operate effectively as additive, path-dependent node costs and that even bounded incentives can induce substantial, network-wide efficiency improvements, supporting the practical applicability of the proposed mechanism.

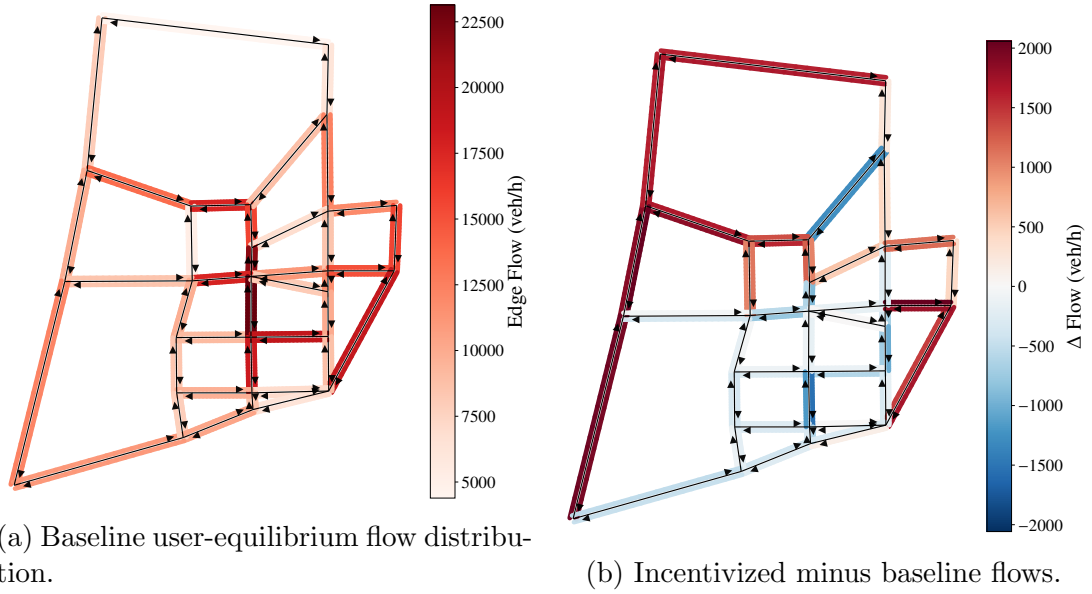


Figure 4.8: Comparison of network flow distributions in the Sioux Falls benchmark. (a) Baseline user-equilibrium (UE) showing congestion concentration on central corridors. (b) Flow redistribution after timestamp-based incentives, where red (and blue) edges depict increased (and decreased) flows.

Chapter 5

Conclusion

Urban traffic congestion is driven not only by limited capacity, but also by the strategic nature of route choice. When each driver minimizes their own travel time, the resulting Wardrop equilibrium need not coincide with the system-optimal flow, and the inefficiency gap can be substantial. While monetary tolling and large-scale infrastructure expansion can address this misalignment in theory, their deployment is often constrained by equity concerns, political feasibility, and cost. This thesis developed an alternative paradigm: using *intersection control* as a non-monetary lever to influence routing outcomes. Intersections are natural coordination points in urban networks, and modern V2I-enabled control architectures allow them to differentiate service in fine-grained ways. By shaping intersection-level delays in a structured manner, it becomes possible to modify perceived path costs and steer selfish routing toward more efficient network-wide flow patterns.

The thesis unified two complementary mechanisms that operationalize this idea at different levels of abstraction. Chapter 3 showed that even simple, locally implementable priority rules can create meaningful path-dependent incentives and substantially improve equilibrium outcomes in canonical networks. Chapter 4 generalized this insight into an analytically tractable framework in which time-stamp modifications induce additive, path-dependent node costs, preserving the

potential-game structure and enabling principled incentive design through bilevel optimization. Together, these two lines of work position intersections not merely as bottlenecks to be mitigated, but as controllable interfaces through which decentralized routing behavior can be guided.

5.1 Main Findings and Contributions

Priority-based incentives (Chapter 3). The first mechanism introduced strategic priority-based scheduling at intersections, where vehicles are treated asymmetrically based on their route choices. The analysis demonstrated that locally administered priority differences can act as effective incentives without relying on monetary transfers. In stylized yet representative networks such as Pigou-type examples, the mechanism shifts equilibrium flows away from overused congestible routes and toward socially preferable allocations. Microscopic SUMO experiments further supported the operational plausibility of the approach and illustrated how small scheduling asymmetries can translate into measurable network-level performance improvements.

Timestamp-based incentives and bilevel design (Chapter 4). The second mechanism extended intersection incentives into a general congestion-game framework by modeling timestamp offsets as additive, path-specific node costs. This formulation preserved key structural properties: the induced game admits a convex potential function, ensuring existence and essential uniqueness (in cost) of equilibrium, and enabling systematic equilibrium computation. The planner’s task naturally becomes a bilevel optimization problem in which incentives are designed with explicit anticipation of selfish equilibrium responses. The framework was grounded in realistic data by calibrating intersection delay functions through SUMO simulations on the Sioux Falls benchmark, and the resulting network-level experiments demonstrated consistent reductions in total travel time. Importantly, the results showed that meaningful efficiency gains remain attainable even under tight bounds on allowable timestamp adjustments.

Unified perspective. Across both mechanisms, the central message is consistent: *intersection-level scheduling choices can be designed as non-monetary, path-dependent incentives that reshape perceived costs and reduce equilibrium inefficiency.* The priority mechanism offers a lightweight, interpretable tool for contexts with limited control sophistication, while the timestamp mechanism provides a scalable, optimization-compatible framework aligned with emerging autonomous intersection technologies.

5.2 Limitations

Several limitations point to directions where the framework can be strengthened. First, the analytical core relies on static, nonatomic routing models. While appropriate for capturing equilibrium structure at the network level, this abstraction does not fully represent dynamic queue evolution, discrete vehicle interactions, or transient demand fluctuations. Second, intersection delays are modeled via smooth, monotone functions calibrated from simulation. Real networks may exhibit non-monotone behavior due to spillback, blocking, and lane-level interactions that couple nodes and links in ways that violate separability. Third, the practical deployment of timestamp-based incentives presumes sufficiently capable V2I communication and controllers that can apply timing adjustments without compromising safety, and such capabilities may be limited in mixed autonomy settings. Finally, bilevel optimization at scale remains computationally demanding; even with simulation-compatible gradient-free methods, large networks can require many equilibrium solves and careful tuning, and real drivers may deviate from perfectly cost-minimizing behavior.

5.3 Future Research Directions

Several research directions naturally follow from this thesis. A first avenue is *mixed autonomy*: understanding how intersection incentives interact with heterogeneous populations of human-driven and autonomous vehicles, including compliance, predictability, and learning effects. A second avenue is *dynamic incentive design*, where timestamp offsets or priority rules adapt online based on real-time congestion estimates and predictive models, potentially yielding improved robustness under time-varying demand. A third direction is the development of *more efficient bilevel algorithms*, leveraging recent advances in differentiable optimization, implicit differentiation for equilibrium problems, or learning-based surrogate models to reduce the number of expensive equilibrium evaluations. A fourth direction is to incorporate *spillback and capacity coupling* explicitly, extending the theory beyond separable node/link models while retaining tractability. Finally, controlled field studies or testbed deployments would help quantify human factors, behavioral responses, and safety-critical constraints under realistic operational conditions.

5.4 Closing Remarks

This thesis showed that intersections can play a dual role in urban mobility: they are sources of delay, but also powerful control points. By treating intersection scheduling decisions as non-monetary incentives, the mechanisms developed here offer a practical pathway for reducing the inefficiency of selfish routing without relying on pricing or major physical changes to infrastructure. The results suggest that as V2I connectivity and autonomous intersection management technologies mature, incentive-aware intersection control can become a viable component of future congestion management systems, enabling cities to achieve measurable efficiency gains while maintaining operational and societal feasibility.

Appendix A

Proof of Proposition 4.9

Consider a non-negative, continuous and non-decreasing function $c(\cdot)$. Then, the mapping

$$x \mapsto \int_0^x c(s) ds \tag{A.1}$$

is continuously differentiable since $\frac{d}{dx} \int_0^x c(s) ds = c(x)$ by the fundamental theorem of calculus. Furthermore, the mapping is convex since

$$\int_x^y c(s) ds \geq c(x)(y - x) = \left(\frac{d}{dx} \int_0^x c(s) ds \right) (y - x).$$

Therefore, Assumption 4.1 yields that the mappings

$$x \mapsto \int_0^x c_e(s) ds \quad \text{and} \quad y \mapsto \int_0^y c_v(s) ds \tag{A.2}$$

are convex and continuously differentiable. Moreover, $f \cdot u$ is linear. This yields that for a given incentive profile u , $\Phi(f, u)$ is a convex and continuously differentiable function of $f \in \mathcal{F}$.

For the equivalence, we prove the following two claims.

Claim A.1. *For a given incentive profile u , if f is an optimal solution of (4.10), then it is an equilibrium flow.*

Proof. Since the objective is convex and the constraints are affine, regularity conditions hold and the Karush–Kuhn–Tucker (KKT) conditions are both necessary and sufficient for optimality. Define the Lagrangian

$$\mathcal{L}(f, \lambda, \mu) = \Phi(f, u) - \lambda \left(\sum_{p \in \mathcal{P}} f_p - 1 \right) - \sum_{p \in \mathcal{P}} f_p \cdot \mu_p. \quad (\text{A.3})$$

Differentiating $\Phi(f, u)$ with respect to f_p , we obtain

$$\frac{\partial \Phi(f, u)}{\partial f_p} = \sum_{e \in p} c_e(f_e) + \sum_{v \in p} c_v^p(f_v, u_v^p) = c_p(f, u). \quad (\text{A.4})$$

Thus, the stationarity condition reads

$$c_p(f, u) - \lambda - \mu_p = 0, \quad \forall p \in \mathcal{P}. \quad (\text{A.5})$$

Complementary slackness implies that if $f_p > 0$, then $\mu_p = 0$ so that

$$c_p(f, u) = \lambda, \quad (\text{A.6})$$

and if $f_p = 0$, then $\mu_p \geq 0$ so that

$$c_p(f, u) = \lambda + \mu_p \geq \lambda. \quad (\text{A.7})$$

Note that (A.6) together with (A.7) are precisely the Wardrop conditions, as described in Definition 4.3. This proves that f is an equilibrium flow. $\square$

Claim A.2. *For a given incentive profile u , if f is an equilibrium flow, then it is an optimal solution of (4.10).*

Proof. Assume that f is an equilibrium flow with equilibrium cost λ , i.e.,

$$c_p(f, u) = \begin{cases} \lambda, & \text{if } f_p > 0, \\ \geq \lambda, & \text{if } f_p = 0. \end{cases} \quad (\text{A.8})$$

Define multipliers by setting $\mu_p = 0$ for $f_p > 0$ and $\mu_p = c_p(f, u) - \lambda \geq 0$ for $f_p = 0$. Then the KKT stationarity condition (A.5) is satisfied. Together with primal feasibility, dual feasibility, and complementary slackness, the KKT conditions hold. Since these conditions are sufficient for optimality in convex problems, it follows that f is an optimal solution. $\square$

Claims A.1 and A.2 complete the proof of Proposition 4.9.

Appendix B

Proof of Theorem 4.10

The flow conservation condition $\mathcal{F}$, as described in (4.1), is a compact subset of $\mathbb{R}^{|\mathcal{P}|}$ and for a given incentive profile u , $\Phi(f, u)$ is continuous on $\mathcal{F}$ by the continuity of $c_e(\cdot)$ and $c_v(\cdot)$. Therefore, the Weierstrass Extreme Value Theorem ensures that $\Phi(\cdot)$ attains its minimum over $\mathcal{F}$ for any given u . To establish essential uniqueness, let f and $\tilde{f}$ be two equilibrium flows associated with u . Since both f and $\tilde{f}$ minimize $\Phi(f)$ by Proposition 4.9 and Φ is convex, we have

$$\begin{aligned}\Phi(f, u) &\leq \Phi(\alpha f + (1 - \alpha)\tilde{f}, u) \\ &\leq \alpha\Phi(f, u) + (1 - \alpha)\Phi(\tilde{f}, u) \\ &= \Phi(f, u)\end{aligned}\tag{B.1}$$

for all $\alpha \in [0, 1]$. The equality for all α ensures that the integrals $\int_0^{f_e} c_e(x) dx$ and $\int_0^{f_v} c_v(y) dy$ are linear between $f_e, \tilde{f}_e$ and $f_v, \tilde{f}_v$, respectively. This implies that their derivatives—corresponding to the marginal costs—are equal:

$$c_e(f_e) = c_e(\tilde{f}_e) \quad \text{and} \quad c_v(f_v) = c_v(\tilde{f}_v)\tag{B.2}$$

for all $e \in E$ and $v \in V$. Then, (4.4) yields that $c_p(f, u) = c_p(\tilde{f}, u) \quad \forall p \in \mathcal{P}$.

Note that the equilibrium condition (4.5) implies that each path with positive flow under f or $\tilde{f}$ has *the same and the smallest* path costs, i.e., for some λ and

$\tilde{\lambda}$, we have

$$\lambda = c_p(f, u) \leq c_{\bar{p}}(f, u) \quad \forall p, \bar{p} \text{ such that } f_p > 0, \quad (\text{B.3a})$$

$$\tilde{\lambda} = c_p(\tilde{f}, u) \leq c_{\bar{p}}(\tilde{f}, u) \quad \forall p, \bar{p} \text{ such that } \tilde{f}_p > 0. \quad (\text{B.3b})$$

Then, (B) yields that $\lambda = \tilde{\lambda}$. The proof is completed since $C(f, u) = \lambda$ and $C(\tilde{f}, u) = \tilde{\lambda}$ by (4.6) and $f \in \mathcal{F}$.

Appendix C

Experimental Setup and Reproducibility

SUMO Scenario Design

Each Sioux Falls intersection is modeled as an independent SUMO scenario that preserves its local geometry, signal timing, and incoming/outgoing lanes. Vehicles are generated according to a Poisson process with flow rate $\lambda = \frac{N}{T}$, where N is the expected number of departures (ranging from 0 to 900) during the simulation horizon $T = 3600$ s and. This captures stochastic arrival behavior typical of real intersections. Vehicle routes are selected uniformly among feasible turn movements. For each vehicle, traversal time is measured from 25 m upstream of the stop line—representing the RSU’s control zone—until complete exit from the intersection, thus capturing both queuing and crossing delays.

Measured travel times are averaged across rollouts (20) to obtain smooth estimates of node traversal cost $c_v(x)$ as a function of flow. Each node’s empirical cost data are then fitted to a quartic model (4.14) using least squares regression. This ensures consistency with the BPR edge-cost formulation (also of fourth order), yielding a unified network-wide congestion representation. Typical MAPE

fall within 1.5–2.5%. Representative fits for Node 10 are shown in Fig. 4.6 and the computed quartic function coefficients are tabulated in Table C.1 for each intersection, illustrated in Fig. 4.5b.

Bilevel Optimization Configuration

The lower-level user equilibrium (UE) computations employ the Frank–Wolfe algorithm with tolerance 10^{-6} and step size $2/(k + 2)$. The upper-level bilevel optimization (planner problem) is solved via the SPSA algorithm using 30,000 iterations and parameters:

$$a = 0.1, \quad \alpha = 0.40, \quad c = 0.4, \quad \gamma = 0.03, \quad A = 1200.$$

Each iteration requires two equilibrium evaluations, resulting in a total of 60,000 UE solves. This setup provides stable convergence while maintaining computational tractability.

All experiments were conducted on a workstation equipped with an Intel® Xeon® w7-3455 CPU and an NVIDIA RTX A4000 GPU (16 GB VRAM). The complete bilevel optimization required approximately two days of wall-clock time.

Table C.1: Fitted quartic node cost coefficients from SUMO simulations for the 24 Sioux Falls intersections. Quartic models follow $c_v(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$. Columns list coefficients (a_0 – a_4) and fitting error (MAPE, %). Rows correspond to the 24 nodes.

Quartic: $y = a_0 + a_1N + a_2N^2 + a_3N^3 + a_4N^4$						
v	a_4	a_3	a_2	a_1	a_0	%
1	-1.14×10^{-14}	3.68×10^{-10}	-2.54×10^{-7}	1.91×10^{-4}	13.99	0.43
2	-3.91×10^{-14}	4.92×10^{-10}	-4.53×10^{-7}	3.36×10^{-4}	13.94	0.42
3	-3.98×10^{-12}	1.96×10^{-8}	-1.56×10^{-5}	5.38×10^{-3}	12.96	1.64
4	-2.81×10^{-12}	1.74×10^{-8}	-1.46×10^{-5}	5.41×10^{-3}	12.91	1.74
5	-4.78×10^{-12}	2.21×10^{-8}	-1.81×10^{-5}	6.23×10^{-3}	12.90	1.69
6	-4.48×10^{-12}	2.09×10^{-8}	-1.68×10^{-5}	5.82×10^{-3}	12.89	1.68
7	-5.75×10^{-14}	5.56×10^{-10}	-5.20×10^{-7}	3.62×10^{-4}	13.93	0.44
8	6.63×10^{-13}	1.02×10^{-8}	-1.06×10^{-5}	5.08×10^{-3}	13.16	1.94
9	-3.35×10^{-12}	1.80×10^{-8}	-1.43×10^{-5}	5.05×10^{-3}	12.98	1.67
10	1.40×10^{-12}	1.26×10^{-8}	-1.45×10^{-5}	6.93×10^{-3}	13.54	2.21
11	1.84×10^{-12}	8.22×10^{-9}	-9.92×10^{-6}	5.16×10^{-3}	13.15	1.83
12	-5.06×10^{-12}	2.23×10^{-8}	-1.78×10^{-5}	6.08×10^{-3}	12.89	1.73
13	6.28×10^{-15}	3.10×10^{-10}	-2.11×10^{-7}	2.16×10^{-4}	13.95	0.45
14	-3.83×10^{-12}	1.91×10^{-8}	-1.53×10^{-5}	5.47×10^{-3}	12.92	1.59
15	4.68×10^{-13}	1.12×10^{-8}	-1.17×10^{-5}	5.39×10^{-3}	13.16	1.90
16	-3.58×10^{-12}	2.24×10^{-8}	-2.15×10^{-5}	8.31×10^{-3}	12.95	2.12
17	-5.66×10^{-12}	2.42×10^{-8}	-1.95×10^{-5}	6.74×10^{-3}	12.87	1.75
18	-5.30×10^{-12}	2.39×10^{-8}	-1.92×10^{-5}	6.50×10^{-3}	13.05	1.74
19	-2.05×10^{-12}	1.53×10^{-8}	-1.27×10^{-5}	4.82×10^{-3}	12.95	1.63
20	-4.90×10^{-12}	3.15×10^{-8}	-2.75×10^{-5}	9.61×10^{-3}	14.50	2.27
21	-5.02×10^{-12}	2.29×10^{-8}	-1.90×10^{-5}	6.70×10^{-3}	12.81	1.76
22	3.01×10^{-12}	4.78×10^{-9}	-6.83×10^{-6}	4.19×10^{-3}	13.39	1.88
23	-3.66×10^{-12}	1.89×10^{-8}	-1.53×10^{-5}	5.44×10^{-3}	12.93	1.68
24	-3.44×10^{-12}	1.88×10^{-8}	-1.55×10^{-5}	5.49×10^{-3}	12.94	1.73

Bibliography

- [1] D. Schrank, B. Eisele, and T. Lomax, “The 2019 urban mobility report,” tech. rep., Texas A&M Transportation Institute with Cooperation from INRIX, 2019.
- [2] T. Litman, *Smart Transportation Emission Reduction Strategies*. Victoria, BC, Canada: Victoria Transport Policy Institute, 2017.
- [3] J. G. Wardrop, “Some theoretical aspects of road traffic research,” *Proceedings of the Institution of Civil Engineers*, vol. 1, no. 3, pp. 325–362, 1952.
- [4] T. Roughgarden, “Routing games,” in *Algorithmic Game Theory*, pp. 461–486, Cambridge: Cambridge University Press, 2007.
- [5] T. Roughgarden and É. Tardos, “How bad is selfish routing?,” *Journal of the ACM*, vol. 49, no. 2, pp. 236–259, 2002.
- [6] A. C. Pigou, *The Economics of Welfare*. London: Macmillan and Co., 1920.
- [7] D. Braess, A. Nagurney, and T. Wakolbinger, “On a paradox of traffic planning,” *Transportation Science*, vol. 39, no. 4, pp. 446–450, 2005.
- [8] L. Fleischer, K. Jain, and M. Mahdian, “Tolls for heterogeneous selfish users in multicommodity networks and generalized congestion games,” in *45th Annual IEEE Symposium on Foundations of Computer Science*, (Rome, Italy), pp. 277–285, IEEE, 2004.
- [9] R. Cole, Y. Dodis, and T. Roughgarden, “Pricing network edges for heterogeneous selfish users,” in *Proceedings of the Thirty-Fifth Annual ACM*

- Symposium on Theory of Computing*, (San Diego, California), pp. 521–530, ACM, 2003.
- [10] Bureau of Public Roads, *Traffic Assignment Manual*. U.S. Department of Commerce, Urban Planning Division, Washington, DC, 1964.
 - [11] M. Beckmann, C. B. McGuire, and C. B. Winsten, *Studies in the Economics of Transportation*. New Haven, Connecticut: Yale University Press, 1956.
 - [12] E. Koutsoupias and C. Papadimitriou, “Worst-case equilibria,” in *Proceedings of the 16th Annual Symposium on Theoretical Aspects of Computer Science* (C. Meinel and S. Tison, eds.), (Berlin, Heidelberg), pp. 404–413, Springer Berlin Heidelberg, 1999.
 - [13] H. Youn, M. T. Gastner, and H. Jeong, “Price of anarchy in transportation networks: efficiency and optimality control,” *Physical Review Letters*, vol. 101, no. 12, p. 128701, 2008.
 - [14] J. Zhang, S. Pourazarm, C. G. Cassandras, and I. C. Paschalidis, “The price of anarchy in transportation networks: Data-driven evaluation and reduction strategies,” *Proceedings of the IEEE*, vol. 106, no. 4, pp. 538–553, 2018.
 - [15] R. Steinberg and W. I. Zangwill, “The prevalence of braess’ paradox,” *Transportation Science*, vol. 17, no. 3, pp. 301–318, 1983.
 - [16] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wießner, “Microscopic traffic simulation using SUMO,” in *Proceedings of the 21st International Conference on Intelligent Transportation Systems (ITSC)*, (Maui, Hawaii), pp. 2575–2582, IEEE, 2018.
 - [17] A. Horni, K. Nagel, and K. W. Axhausen, *The Multi-Agent Transport Simulation MATSim*. London: Ubiquity Press, 2016.
 - [18] A. A. Walters, “The theory and measurement of private and social cost of highway congestion,” *Econometrica*, vol. 29, no. 4, pp. 676–699, 1961.

- [19] H. Yang and H.-J. Huang, “Principle of marginal-cost pricing: How does it work in a general road network?,” *Transportation Research Part A: Policy and Practice*, vol. 32, no. 1, pp. 45–54, 1998.
- [20] T. Akamatsu, “Decomposition of path choice entropy in general transport networks,” *Transportation Science*, vol. 31, no. 4, pp. 349–362, 1997.
- [21] J. Eliasson, “Is congestion pricing fair? consumer and citizen perspectives on equity effects,” *Transport Policy*, vol. 52, pp. 1–15, 2016.
- [22] T. Litman, *Evaluating Transportation Equity*. Victoria, BC, Canada: Victoria Transport Policy Institute, 2017.
- [23] J. Eliasson, “The stockholm congestion charges: an overview,” CTS Working Paper 2014:7, Centre for Transport Studies, Stockholm, Sweden, 2014.
- [24] T. Litman, *Transportation Cost and Benefit Analysis*. Victoria, BC, Canada: Victoria Transport Policy Institute, 2009.
- [25] P. Rezaeinia, A. de Palma, and R. Lindsey, “Traffic congestion control: Tradable permits vs road tolls,” working paper, THEMA, CY Cergy Paris Université, France, 2021.
- [26] H. Yang and X. Wang, “Managing network mobility with tradable credits,” *Transportation Research Part B: Methodological*, vol. 45, no. 3, pp. 580–594, 2011.
- [27] A. Censi, S. Bolognani, J. G. Zilly, S. S. Mousavi, and E. Frazzoli, “Today me, tomorrow thee: Efficient resource allocation in competitive settings using Karma games,” in *Proceedings of the 22nd IEEE International Conference on Intelligent Transportation Systems (ITSC)*, (Auckland, New Zealand), pp. 686–693, IEEE, 2019.
- [28] K. Gemici, E. Koutsoupias, B. Monnot, C. Papadimitriou, and G. Piliouras, “Wealth inequality and the price of anarchy,” in *36th International Symposium on Theoretical Aspects of Computer Science (STACS 2019)*, vol. 126 of *LIPICs*, pp. 31:1–31:16, Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, 2019.

- [29] E. Kamenica and M. Gentzkow, “Bayesian persuasion,” *American Economic Review*, vol. 101, no. 6, pp. 2590–2615, 2011.
- [30] S. Das, E. Kamenica, and R. Mirka, “Reducing congestion through information design,” in *55th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, (Monticello, Illinois), pp. 1279–1284, IEEE, 2017.
- [31] H. Tavafoghi and D. Teneketzis, “Strategic information provision in routing games,” *IEEE Transactions on Control of Network Systems*, vol. 6, no. 4, pp. 1267–1278, 2019.
- [32] H. Tavafoghi, A. Shetty, K. Poolla, and P. Varaiya, “Strategic information platforms in transportation networks,” in *57th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, (Monticello, Illinois), pp. 816–823, IEEE, 2019.
- [33] C. Siripanpornchana, S. Panichpapiboon, and P. Chaovalit, “Travel-time prediction with deep learning,” in *IEEE Region 10 Conference (TENCON)*, pp. 1859–1862, IEEE, 2016.
- [34] X. Li, G. Cong, A. Sun, and Y. Cheng, “Learning travel time distributions with deep generative model,” in *Proceedings of the World Wide Web Conference (WWW)*, pp. 1017–1027, 2019.
- [35] Z.-C. Li, Z.-K. Li, and W. H. K. Lam, “An integrated design of sustainable land use and transportation system with uncertainty in future population,” *Transportmetrica A: Transport Science*, vol. 10, no. 2, pp. 160–185, 2014.
- [36] T. Roughgarden, *Selfish Routing and the Price of Anarchy*. MIT Press, 2005.
- [37] M. G. H. Bell and Y. Iida, *Transportation Network Analysis*. John Wiley & Sons, 1997.
- [38] T. L. Magnanti and R. T. Wong, “Network design and transportation planning: Models and algorithms,” *Transportation Science*, vol. 18, no. 1, pp. 1–55, 1984.

- [39] T. Roughgarden, “Designing networks for selfish users is hard,” in *Proceedings of the 42nd Annual Symposium on Foundations of Computer Science*, pp. 472–481, IEEE, 2001.
- [40] N. Stamatiadis and A. Kirk, “Guidelines for road diet conversions,” in *Proceedings of the 9th Asia Pacific Conference on Transportation and the Environment (APTE)*, p. 358, 2014.
- [41] P. Wu, F. Chu, and A. Che, “Mixed-integer programming for a new bus-lane reservation problem,” in *Proceedings of the 18th IEEE International Conference on Intelligent Transportation Systems (ITSC)*, (Gran Canaria, Spain), pp. 2782–2787, IEEE, 2015.
- [42] O. Jahn, R. H. Möhring, A. S. Schulz, and N. E. Stier-Moses, “System-optimal routing of traffic flows with user constraints in networks with congestion,” *Operations Research*, vol. 53, no. 4, pp. 600–616, 2005.
- [43] Q. Meng and H. Yang, “Benefit distribution and equity in road network design,” *Transportation Research Part B: Methodological*, vol. 36, no. 1, pp. 19–35, 2002.
- [44] S. Lawphongpanich and Y. Yin, “Solving the Pareto-improving toll problem via manifold suboptimization,” *Transportation Research Part C: Emerging Technologies*, vol. 18, no. 2, pp. 234–246, 2010.
- [45] M. Papageorgiou, C. Diakaki, V. Dinopoulou, A. Kotsialos, and Y. Wang, “Review of road traffic control strategies,” *Proceedings of the IEEE*, vol. 91, no. 12, pp. 2043–2067, 2003.
- [46] K. Dresner and P. Stone, “A multiagent approach to autonomous intersection management,” *Journal of Artificial Intelligence Research*, vol. 31, pp. 591–656, 2008.
- [47] Y.-T. Lin, H. Hsu, S.-C. Lin, C.-W. Lin, I. H.-R. Jiang, and C.-W. Liu, “Graph-based modeling, scheduling, and verification for intersection management of intelligent vehicles,” *ACM Transactions on Embedded Computing Systems*, vol. 18, no. 5s, pp. 1–21, 2019.

- [48] Y. Zhang and C. G. Cassandras, “Decentralized optimal control of connected automated vehicles at signal-free intersections including comfort-constrained turns and safety guarantees,” *Automatica*, vol. 109, p. 108563, 2019.
- [49] T. Harks, B. Peis, D. Schmand, B. Tauer, and L. Vargas Koch, “Competitive packet routing with priority lists,” *ACM Transactions on Economics and Computation*, vol. 6, no. 1, pp. 4:1–4:26, 2018.
- [50] Z. Zhong, M. Nejad, and E. E. Lee, “Autonomous and semiautonomous intersection management: A survey,” *IEEE Intelligent Transportation Systems Magazine*, vol. 13, no. 2, pp. 53–70, 2021.
- [51] N. H. Gartner, C. J. Messer, and A. K. Rath, “Traffic flow theory: A state-of-the-art report,” revised monograph, Federal Highway Administration, Washington, DC, 2001.
- [52] National Academies of Sciences, Engineering, and Medicine, *Highway Capacity Manual 7th Edition: A Guide for Multimodal Mobility Analysis*. Washington, DC: The National Academies Press, 2022.
- [53] F. V. Webster, “Traffic signal settings,” Road Research Technical Paper 39, Road Research Laboratory, London, UK, 1958.
- [54] R. E. Allsop, “Estimating the traffic capacity of a signalized road junction,” *Transportation Research*, vol. 6, no. 3, pp. 245–255, 1972.
- [55] D. I. Robertson, “TRANSYT: A traffic network study tool,” RRL Report LR 253, Road Research Laboratory, Crowthorne, UK, 1969.
- [56] D. I. Robertson and R. D. Bretherton, “Optimizing networks of traffic signals in real time—the SCOOT method,” *IEEE Transactions on Vehicular Technology*, vol. 40, no. 1, pp. 11–15, 1991.
- [57] P. R. Lowrie, “SCATS, sydney co-ordinated adaptive traffic system: A traffic responsive method of controlling urban traffic,” tech. rep., Roads and Traffic Authority of New South Wales, Sydney, Australia, 1990.

- [58] P. Mirchandani and L. Head, “A real-time traffic signal control system: Architecture, algorithms, and analysis,” *Transportation Research Part C: Emerging Technologies*, vol. 9, no. 6, pp. 415–432, 2001.
- [59] W. R. McShane and R. P. Roess, *Traffic Engineering*. Englewood Cliffs, NJ: Prentice Hall, 1990.
- [60] T. Wongpiromsarn, T. Uthaicharoenpong, Y. Wang, E. Frazzoli, and D. Wang, “Distributed traffic signal control for maximum network throughput,” in *Proceedings of the 15th IEEE International Conference on Intelligent Transportation Systems (ITSC)*, (Anchorage, AK, USA), pp. 588–595, IEEE, 2012.
- [61] P. Varaiya, “Max pressure control of a network of signalized intersections,” *Transportation Research Part C: Emerging Technologies*, vol. 36, pp. 177–195, 2013.
- [62] K. Aboudolas, M. Papageorgiou, and E. Kosmatopoulos, “Store-and-forward based methods for the signal control problem in large-scale congested urban road networks,” *Transportation Research Part C: Emerging Technologies*, vol. 17, no. 2, pp. 163–174, 2009.
- [63] S. Lin, B. De Schutter, Y. Xi, and H. Hellendoorn, “Fast model predictive control for urban road networks via MILP,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 12, no. 3, pp. 846–856, 2011.
- [64] R. Akcelik, “Traffic signals: Capacity and timing analysis,” ARR Report 123, Australian Road Research Board, Vermont South, Victoria, 1981.
- [65] A. J. Miller, “A queueing model for road traffic flow,” *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 23, no. 1, pp. 64–75, 1961.
- [66] K. Zhang, D. Zhang, A. de La Fortelle, X. Wu, and J. Gregoire, “State-driven priority scheduling mechanisms for driverless vehicles approaching intersections,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 16, no. 5, pp. 2487–2500, 2015.

- [67] M. Ferreira and P. M. d'Orey, "On the impact of virtual traffic lights on carbon emissions mitigation," *IEEE Transactions on Intelligent Transportation Systems*, vol. 13, no. 1, pp. 284–295, 2012.
- [68] M. Khayatian, M. Mehrabian, E. Andert, R. Dedinsky, S. Choudhary, Y. Lou, and A. Shirvastava, "A survey on intersection management of connected autonomous vehicles," *ACM Transactions on Cyber-Physical Systems*, vol. 4, no. 4, pp. 47:1–47:27, 2020.
- [69] J. Rios-Torres and A. A. Malikopoulos, "A survey on the coordination of connected and automated vehicles at intersections and merging at highway on-ramps," *IEEE Transactions on Intelligent Transportation Systems*, vol. 18, no. 5, pp. 1066–1077, 2017.
- [70] J. B. Kenney, "Dedicated short-range communications (DSRC) standards in the United States," *Proceedings of the IEEE*, vol. 99, no. 7, pp. 1162–1182, 2011.
- [71] R. Weber, J. Misener, and V. Park, "C-V2X – a communication technology for cooperative, connected and automated mobility," in *Proceedings of the 24th ITG-Symposium on Mobile Communication-Technologies and Applications*, (Osnabrueck, Germany), pp. 1–6, VDE, 2019.
- [72] D. Fajardo, T.-C. Au, S. T. Waller, P. Stone, and D. Yang, "Automated intersection control: Performance of future innovation versus current traffic signal control," *Transportation Research Record: Journal of the Transportation Research Board*, vol. 2259, no. 1, pp. 223–232, 2011.
- [73] M. W. Levin and D. Rey, "Conflict-point formulation of intersection control for autonomous vehicles," *Transportation Research Part C: Emerging Technologies*, vol. 85, pp. 528–547, 2017.
- [74] M. Bashiri, H. Jafarzadeh, and C. H. Fleming, "PAIM: Platoon-based autonomous intersection management," in *Proceedings of the 21st International Conference on Intelligent Transportation Systems (ITSC)*, (Maui, Hawaii), pp. 374–380, IEEE, 2018.

- [75] B. Xu, S. E. Li, Y. Bian, S. Li, X. J. Ban, J. Wang, and K. Li, “Distributed conflict-free cooperation for multiple connected vehicles at unsignalized intersections,” *Transportation Research Part C: Emerging Technologies*, vol. 93, pp. 322–334, 2018.
- [76] S. R. Azimi, G. Bhatia, R. Rajkumar, and P. Mudalige, “Reliable intersection protocols using vehicular networks,” in *Proceedings of the 4th ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)*, (Philadelphia, PA, USA), pp. 1–10, 2013.
- [77] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, “Control barrier function based quadratic programs for safety critical systems,” *IEEE Transactions on Automatic Control*, vol. 62, no. 8, pp. 3861–3876, 2017.
- [78] M. Althoff, O. Stursberg, and M. Buss, “Computing reachable sets of hybrid systems using a combination of zonotopes and polytopes,” *Nonlinear Analysis: Hybrid Systems*, vol. 4, no. 2, pp. 233–249, 2010.
- [79] A. Colombo and D. Del Vecchio, “Least restrictive supervisors for intersection collision avoidance: A scheduling approach,” *IEEE Transactions on Automatic Control*, vol. 60, no. 6, pp. 1515–1527, 2015.
- [80] D. Isele, R. Rahimi, A. Cosgun, K. Subramanian, and K. Fujimura, “Navigating occluded intersections with autonomous vehicles using deep reinforcement learning,” in *Proceedings of the 2018 IEEE International Conference on Robotics and Automation (ICRA)*, (Brisbane, QLD, Australia), pp. 2034–2039, IEEE, 2018.
- [81] B. R. Kiran, I. Sobh, V. Talpaert, P. Mannion, A. A. Al Sallab, S. Yogamani, and P. Pérez, “Deep reinforcement learning for autonomous driving: A survey,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 6, pp. 4909–4926, 2022.
- [82] T. Chu, J. Wang, L. Codecà, and Z. Li, “Multi-agent deep reinforcement learning for large-scale traffic signal control,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 3, pp. 1086–1095, 2020.

- [83] A. E. Sallab, M. Abdou, E. Perot, and S. Yogamani, “Deep reinforcement learning framework for autonomous driving,” in *Proceedings of the IS&T International Symposium on Electronic Imaging*, vol. 2017, pp. 70–76, Society for Imaging Science and Technology, 2017.
- [84] A. Talebpour and H. S. Mahmassani, “Influence of connected and autonomous vehicles on traffic flow stability and throughput,” *Transportation Research Part C: Emerging Technologies*, vol. 71, pp. 143–163, 2016.
- [85] R. Hassin and M. Haviv, *To Queue or Not to Queue: Equilibrium Behavior in Queueing Systems*, vol. 59 of *International Series in Operations Research & Management Science*. Boston, MA: Springer, 2003.
- [86] W. Vickrey, “Congestion theory and transport investment,” *American Economic Review*, vol. 59, no. 2, pp. 251–260, 1969.
- [87] P. Naor, “The regulation of queue size by levying tolls,” *Econometrica*, vol. 37, no. 1, pp. 15–24, 1969.
- [88] M. W. Levin and S. D. Boyles, “A multiclass cell transmission model for shared human and autonomous vehicle roads,” *Transportation Research Part C: Emerging Technologies*, vol. 62, pp. 103–116, 2016.
- [89] M. W. Levin and S. D. Boyles, “Intersection auctions and reservation-based control in dynamic traffic assignment,” *Transportation Research Record: Journal of the Transportation Research Board*, vol. 2497, no. 1, pp. 35–44, 2015.
- [90] M. Vasirani and S. Ossowski, “A market-inspired approach to reservation-based urban road traffic management,” in *Proceedings of the 8th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pp. 617–624, 2009.
- [91] Q. Jin, G. Wu, K. Boriboonsomsin, and M. Barth, “Platoon-based multi-agent intersection management for connected vehicle,” in *Proceedings of the 16th IEEE International Conference on Intelligent Transportation Systems (ITSC)*, (The Hague, Netherlands), pp. 1462–1467, IEEE, 2013.

- [92] R. Tachet, P. Santi, S. Sobolevsky, L. I. Reyes-Castro, E. Frazzoli, D. Helbing, and C. Ratti, “Revisiting street intersections using slot-based systems,” *PLOS ONE*, vol. 11, no. 3, p. e0149607, 2016.
- [93] M. Wu, L. Jin, S. Amin, and P. Jaillet, “Signaling game-based misbehavior inspection in V2I-enabled highway operations,” in *Proceedings of the 57th IEEE Conference on Decision and Control (CDC)*, (Miami Beach, FL, USA), pp. 2728–2734, IEEE, 2018.
- [94] E. Elokda, S. Bolognani, A. Censi, F. Dörfler, and E. Frazzoli, “A self-contained karma economy for the dynamic allocation of common resources,” *Dynamic Games and Applications*, vol. 14, no. 3, pp. 578–610, 2024.
- [95] C. F. Daganzo, “The cell transmission model, part II: Network traffic,” *Transportation Research Part B: Methodological*, vol. 29, no. 2, pp. 79–93, 1995.
- [96] W. Y. Szeto and H. K. Lo, “Dynamic traffic assignment: Properties and extensions,” *Transportmetrica*, vol. 2, no. 1, pp. 31–52, 2006.
- [97] L. Hurwicz, “On informationally decentralized systems,” in *Decision and Organization: A Volume in Honor of Jacob Marschak* (C. B. McGuire and R. Radner, eds.), pp. 297–336, Amsterdam: North-Holland, 1972.
- [98] E. Maskin, “Nash equilibrium and welfare optimality,” *The Review of Economic Studies*, vol. 66, no. 1, pp. 23–38, 1999.
- [99] D. Carlino, S. D. Boyles, and P. Stone, “Auction-based autonomous intersection management,” in *16th International IEEE Conference on Intelligent Transportation Systems (ITSC 2013)*, pp. 529–534, IEEE, 2013.
- [100] Y. Saltan, A. Kosay, C.-W. Lin, and M. O. Sayin, “Strategic control of intersections for efficient traffic routing without tolls,” in *Proceedings of the 5th IFAC Workshop on Cyber-Physical Human Systems*, vol. 58, (Antalya, Türkiye), pp. 175–180, IFAC, 2024.

- [101] Y. Saltan, J.-J. Wang, A. Kosay, C.-W. Lin, and M. O. Sayin, “Designing non-monetary intersection control mechanisms for efficient selfish routing,” *arXiv preprint arXiv:2511.01421*, 2025.
- [102] Y. Guan, Y. Ren, S. E. Li, Q. Sun, L. Luo, , and K. Li, ““centralized cooperation for connected and automated vehicles at intersections by proximal policy optimization,” *IEEE Transactions on Vehicular Technology*, vol. 69, pp. 12 597–12 608, 2020.
- [103] Y. Wu, H. Chen, and F. Zhu, “DCL-AIM: Decentralized coordination learning of autonomous intersection management for connected and automated vehicles,” *Transportation Research Part C: Emerging Technologies*, vol. 103, pp. 246–260, 2019.
- [104] S.-C. Huang, K.-E. Lin, C.-Y. Kuo, L.-H. Lin, M. O. Sayin, and C.-W. Lin, “Reinforcement-learning-based job-shop scheduling for intelligent intersection management,” in *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, (Antwerp, Belgium), pp. 1–6, IEEE, 2023.
- [105] I. Milchtaich, “Congestion games with player-specific payoff functions,” *Games and Economic Behavior*, vol. 13, pp. 111–124, 1996.
- [106] I. Milchtaich, “Weighted congestion games with separable preferences,” *Games and Economic Behavior*, vol. 67, pp. 750–757, 2009.
- [107] R. Cole, Y. Dodis, and T. Roughgarden, “How much can taxes help selfish routing?,” *Journal of Computer and System Sciences*, vol. 72, no. 3, pp. 444–467, 2006.
- [108] T. Harks and M. Klimm, “On the existence of pure Nash equilibria in weighted congestion games,” *Mathematics of Operations Research*, vol. 37, no. 3, pp. 419–436, 2012.
- [109] M. J. Smith, “The existence, uniqueness and stability of traffic equilibria,” *Transportation Research Part B: Methodological*, vol. 13, no. 4, pp. 295–304, 1979.

- [110] S. C. Endres, C. Sandrock, and W. W. Focke, “A simplicial homology algorithm for Lipschitz optimisation,” *Journal of Global Optimization*, vol. 72, pp. 181–217, 2018.
- [111] R. Storn and K. Price, “Differential evolution - A simple and efficient heuristic for global optimization over continuous spaces,” *Journal of Global Optimization*, vol. 11, pp. 341–359, 1997.
- [112] Y. Xiang, D. Y. Sun, W. Fan, and X. G. Gong, “Generalized simulated annealing algorithm and its application to the Thomson model,” *Physics Letters A*, vol. 233, no. 3, pp. 216–220, 1997.
- [113] L. J. LeBlanc, E. K. Morlok, and W. P. Pierskalla, “An efficient approach to solving the road network equilibrium traffic assignment problem,” *Transportation Research*, vol. 9, no. 5, pp. 309–318, 1975.
- [114] J. C. Spall, “Multivariate stochastic approximation using a simultaneous perturbation gradient approximation,” *IEEE Transactions on Automatic Control*, vol. 37, no. 3, pp. 332–341, 1992.